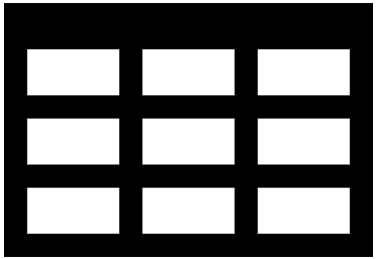
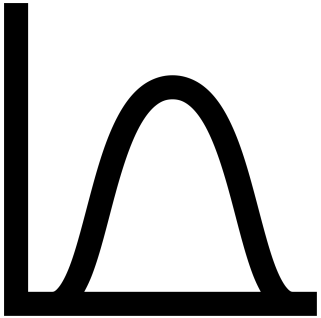
Distributions & Linear Models

There are many metrics by which one can describe a dataset.

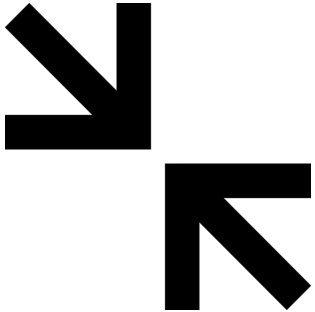


Size

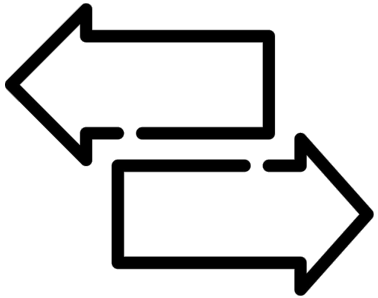
Missingness



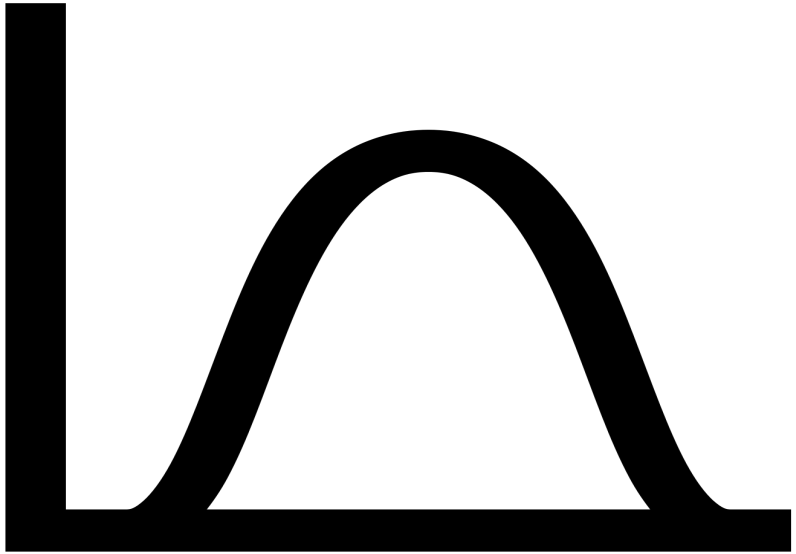
Shape



Central Tendency



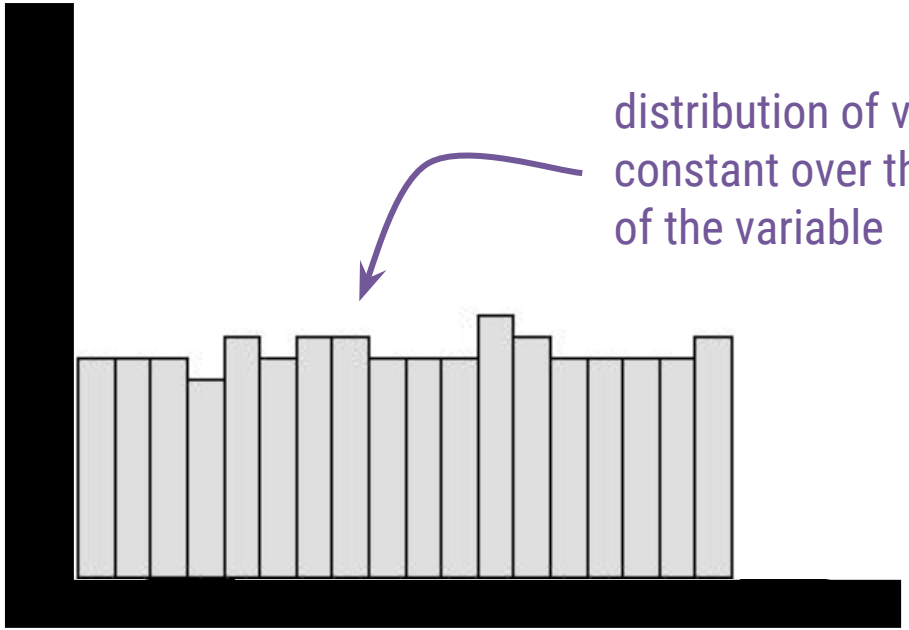
Variability



Shape

Here, we'll focus on how **shape** dictates your analytic approach.

A Uniform Distribution



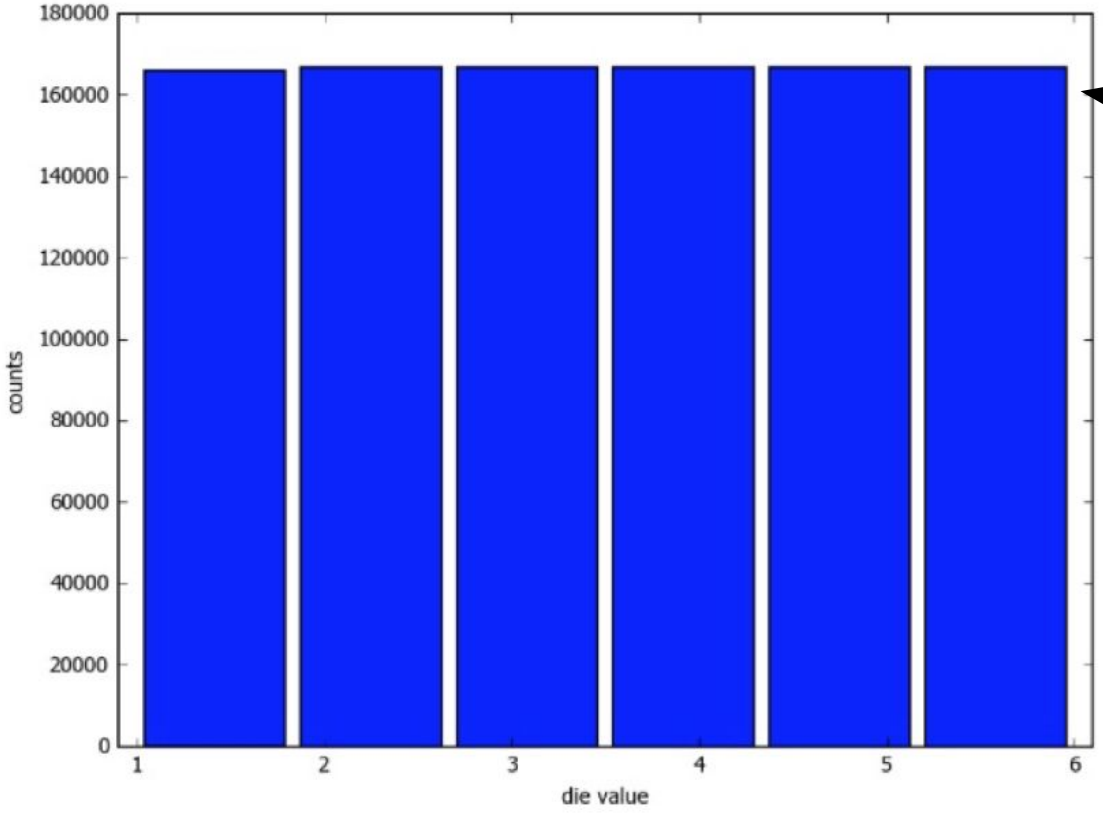
Occurs when every possible outcome has an *equally-likely chance* of occurring

All distribution images adapted from: https://www.pqsystems.com/qualityadvisor/DataAnalysisTools/interpretation/histogram_shape.php

```python
num_rolls = 1000000

simulated_rolls = np.random.randint(low=1, high=7, size=num_rolls)

# np.unique finds the unique values are in our data (1, 2, 3, 4, 5, 6)
# and len asks how many there are (6)
# we then use that to define the number of bins we want in our histogram
die_vals = np.unique(simulated_rolls)
number_of_bins = len(die_vals)
spacing = 0.1

plt.hist(simulated_rolls, bins=number_of_bins, rwidth=1-spacing)
plt.xlim(np.min(die_vals)-spacing, np.max(die_vals)+spacing)
plt.xlabel('die value')
plt.ylabel('counts')
plt.show()
```

What if we rolled a die 1M times (by simulation)?

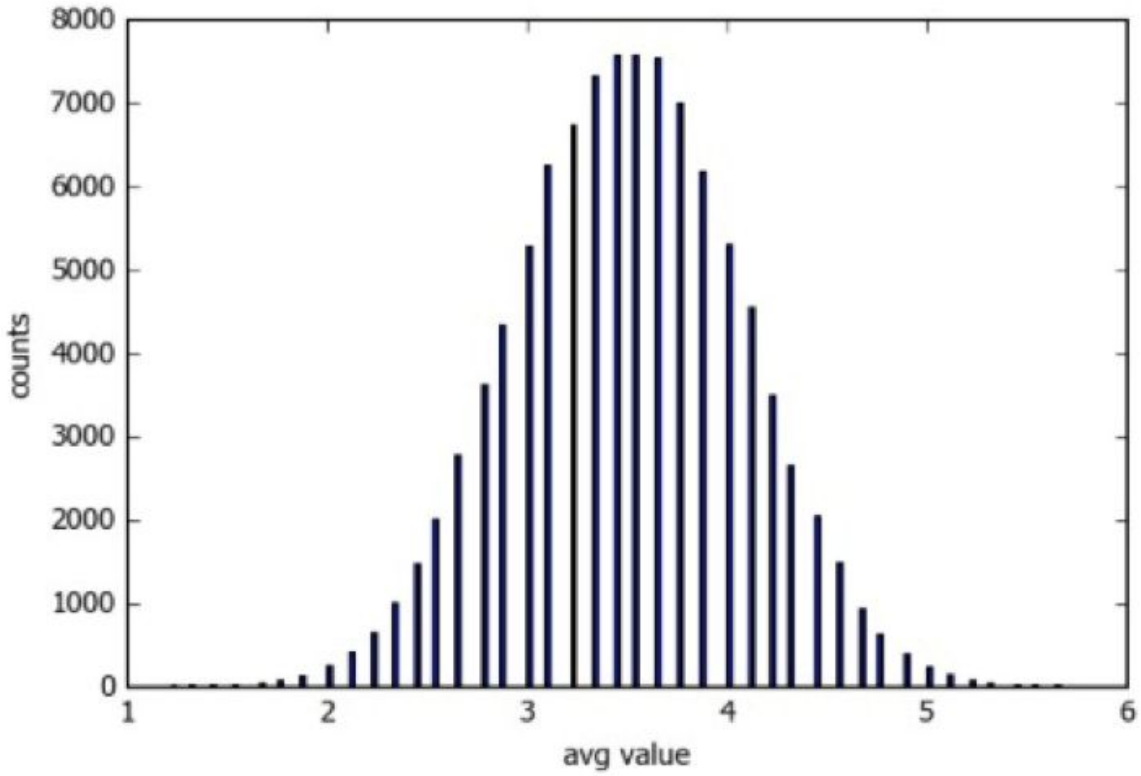
Distribution is pretty **uniform** (aka flat)



But, what if we take our die rolling results (`simulated_rolls`) and break those into chunks of every 10 rolls...

If we take the average of each of those 10 rolls...

```
chunk_size = 10 # each chunk has this many rolls
samples = num_rolls//chunk_size # instead of <num_rolls> elements, we now have num_rolls/10
                                # so instead of 1,000,000 we have 100,000
roll_avg = np.zeros((samples, 1)) # initialize your vector

for i in range(samples):
    # for each loop we need to average from 0:9, 10:19, 20:29, and so on
    # this gives us a low range and a high range to average across
    # the low range (0,10,20,etc) is just i*10
    # the high range is that, plus 9
    low_range = i*10
    high_range = low_range+9
    roll_avg[i] = np.mean(simulated_rolls[low_range:high_range]) # average the chunks

# plot the histogram!
plt.hist(roll_avg, 200)
plt.xlabel('avg value')
plt.ylabel('counts')
plt.show()

print(np.mean((1,2,3,4,5,6)), np.mean(roll_avg))
```

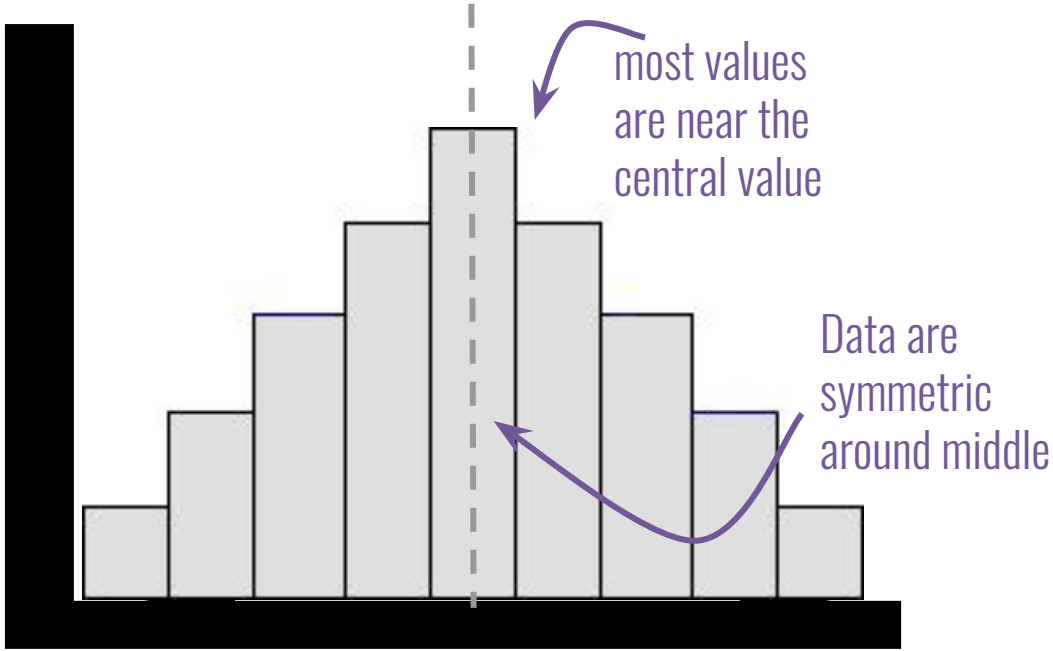


The distribution of these averages is **Normally** distributed!

```
3.5 3.49934888889
```

We'll return to discussing this (the Central Limit Theorem) in a later lecture.

A Normal Distribution

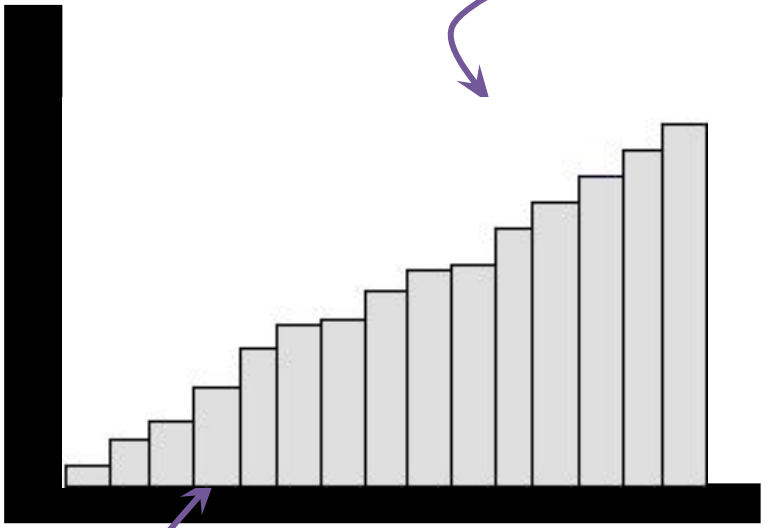


...aka Gaussian or bell-shaped Distribution

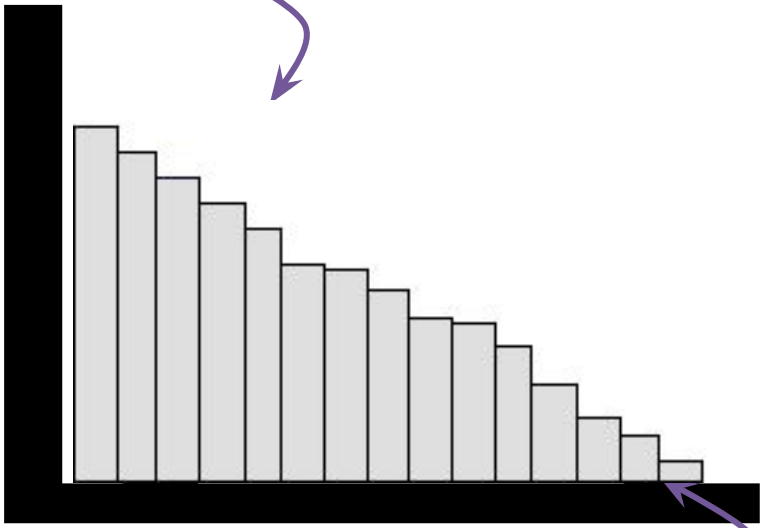
All distribution images adapted from: https://www.pqsystems.com/qualityadvisor/DataAnalysisTools/interpretation/histogram_shape.php

A Skewed Distribution

most values fall to one extreme within the range



skewed left



skewed right

All distribution images adapted from: https://www.pqsystems.com/qualityadvisor/DataAnalysisTools/interpretation/histogram_shape.php

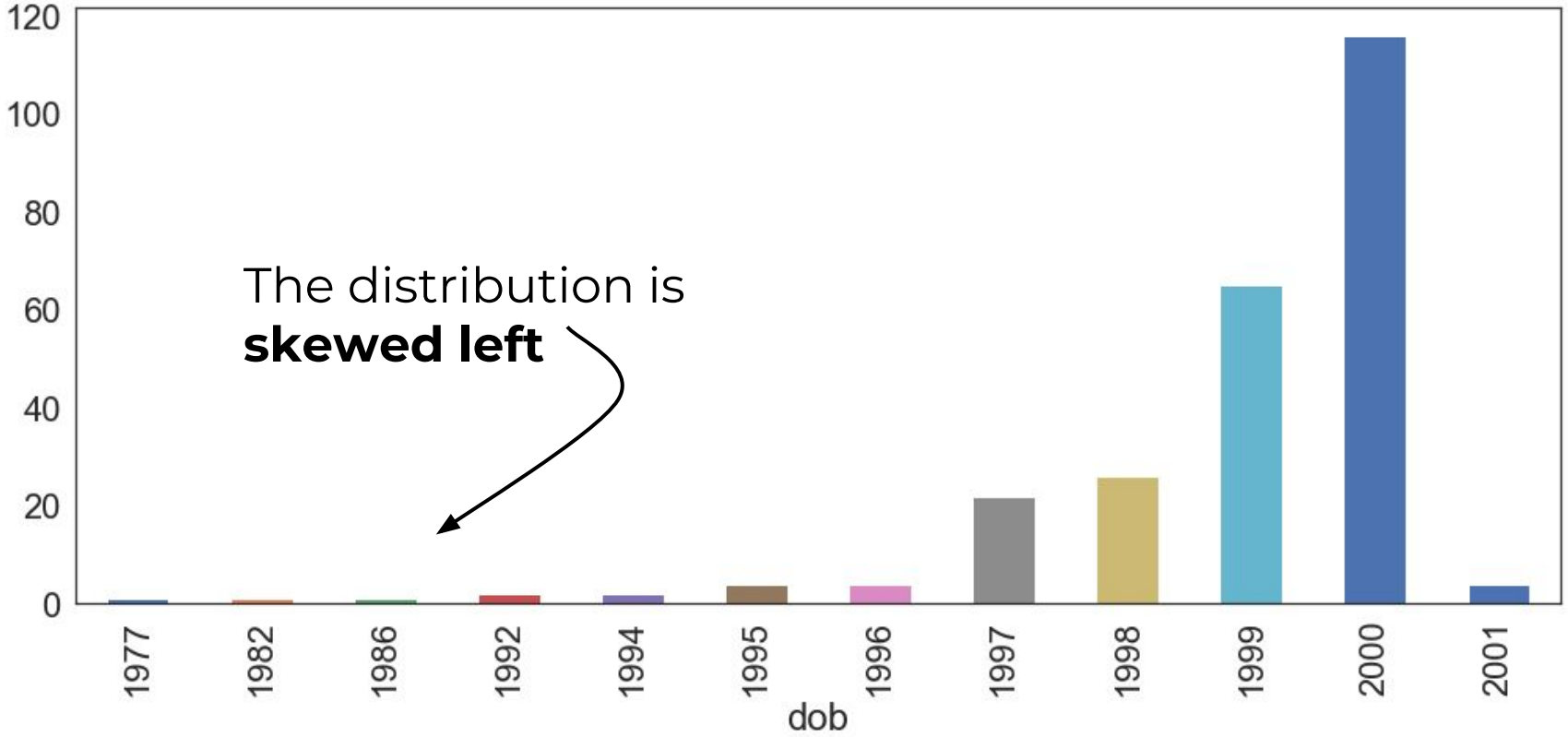
What if we were to look at some class data from UCSD students?

Dataset contains height, date of birth, and background experience in statistics and programming

	height	dob	statistics	programming
0	65.0	1999-04-08	2	9
1	63.0	1997-10-28	1	3
2	66.0	2000-05-29	7	3
3	63.0	2000-05-11	8	1
4	67.0	1995-02-01	1	10
5	72.0	2001-03-05	8	6
6	71.0	1997-09-30	7	5
7	65.0	1999-04-08	2	9
8	68.0	2000-07-30	10	8
9	67.0	2000-08-19	6	4

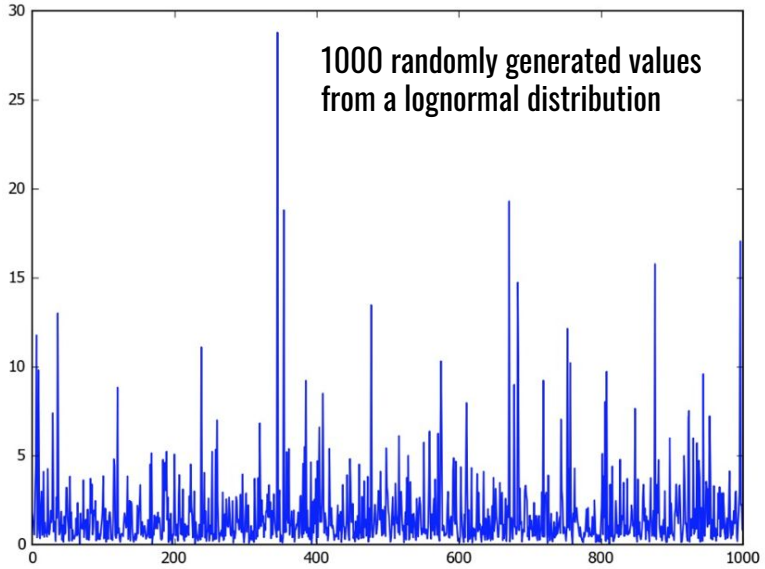
What would the distribution of students' ages look like?

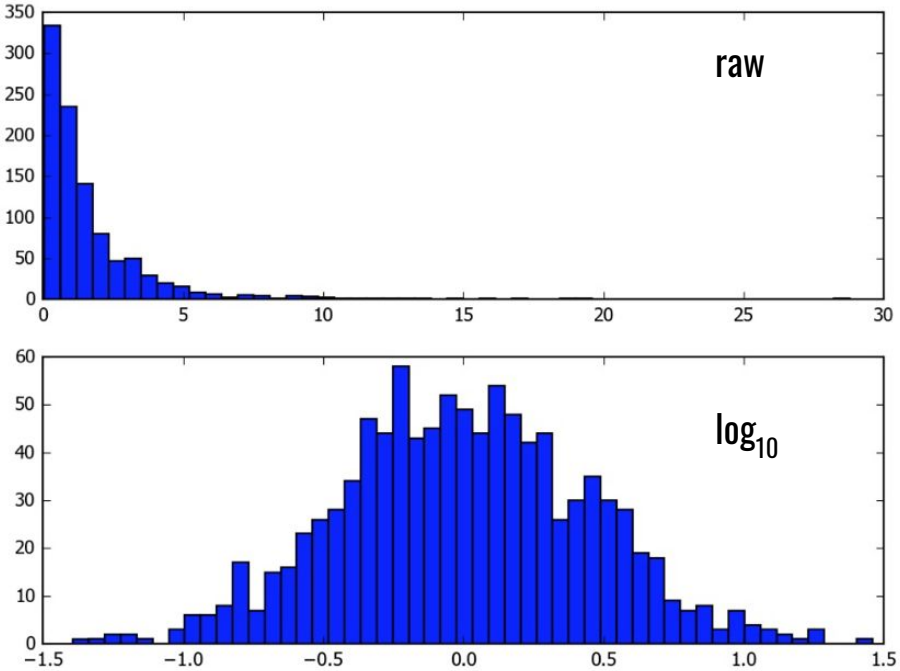
```
import pandas as pd

import matplotlib as mpl
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = (17, 7)

import seaborn as sns
sns.set(font_scale=2,style="white")

import datetime

# read data in
df = pd.read_csv('https://raw.githubusercontent.com/shanellis/datasets/master/student_data.csv')

# edit datetime
df['dob'] = pd.to_datetime(df['dob'])
# filter out those who misread the question
df = df[df['dob'] < '2018-01-01']

# plot DOB data
df['dob'].groupby(df["dob"].dt.year).count().plot(kind="bar")
```


120
100
80
60
40
20
0
The distribution is
skewed left
1977
1982
1986
1992
1994
1995
1996
1997
1998
1999
2000
2001
dob

Lognormal

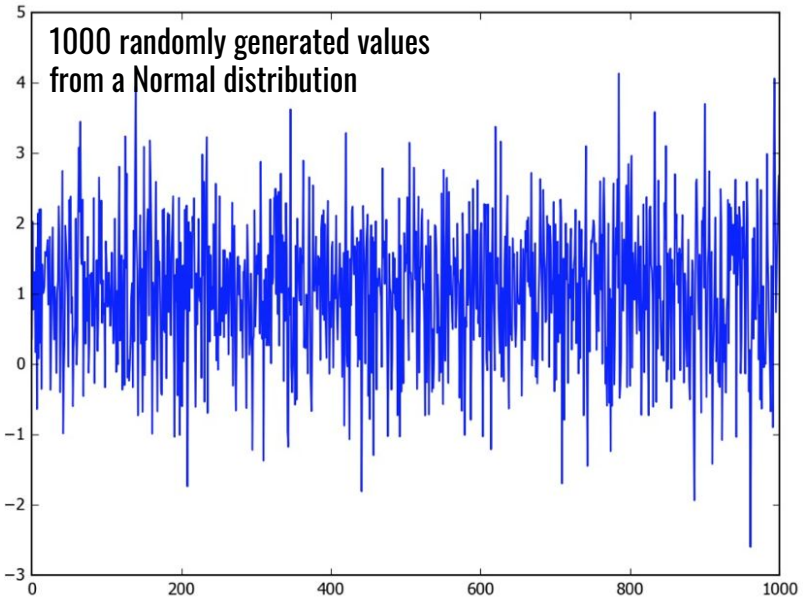
distributions are skewed distributions with a long right tail

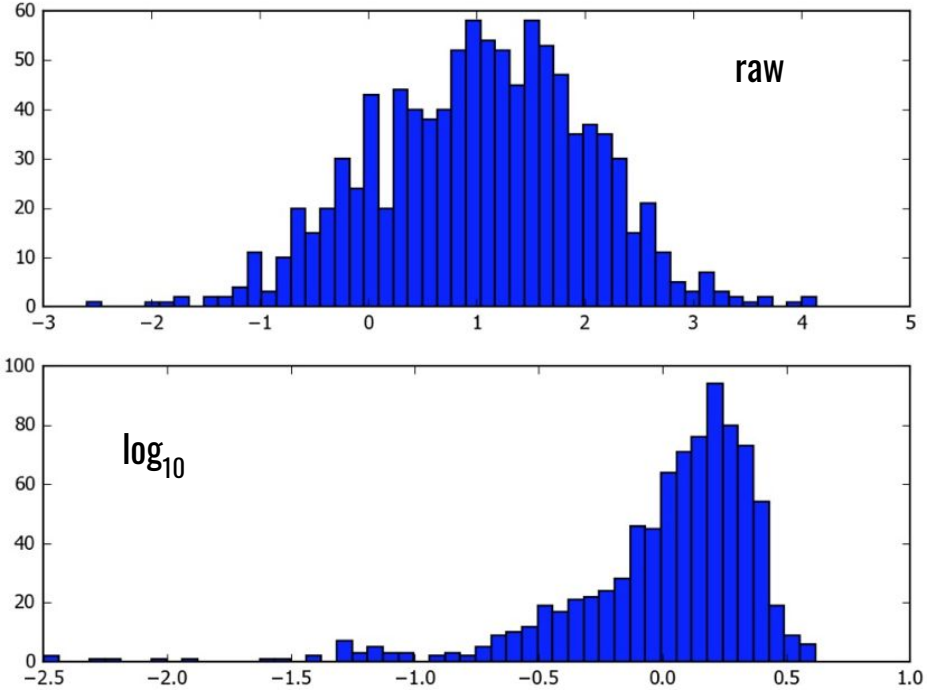




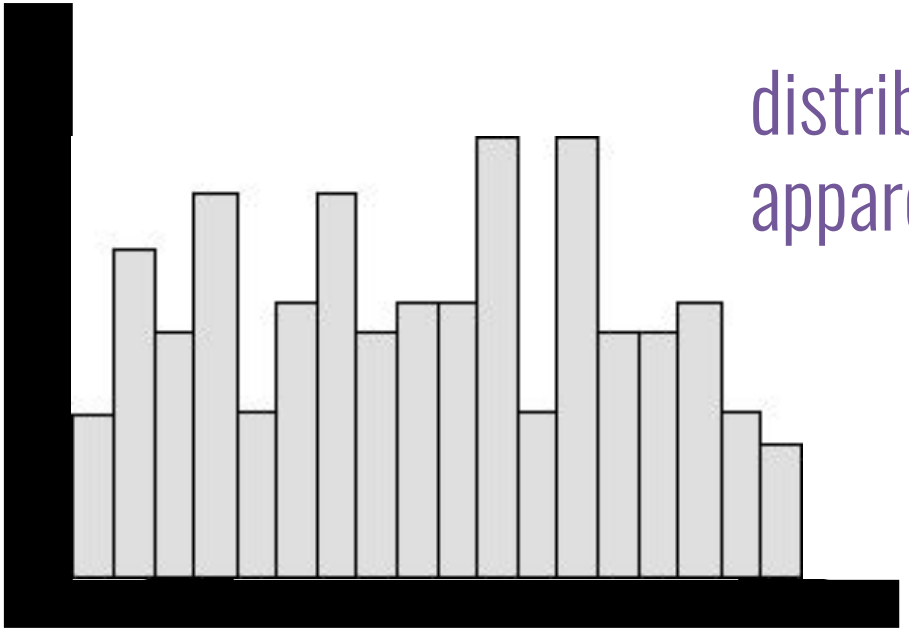
Normal

Log-transformed values from the Normal distribution are skewed-left





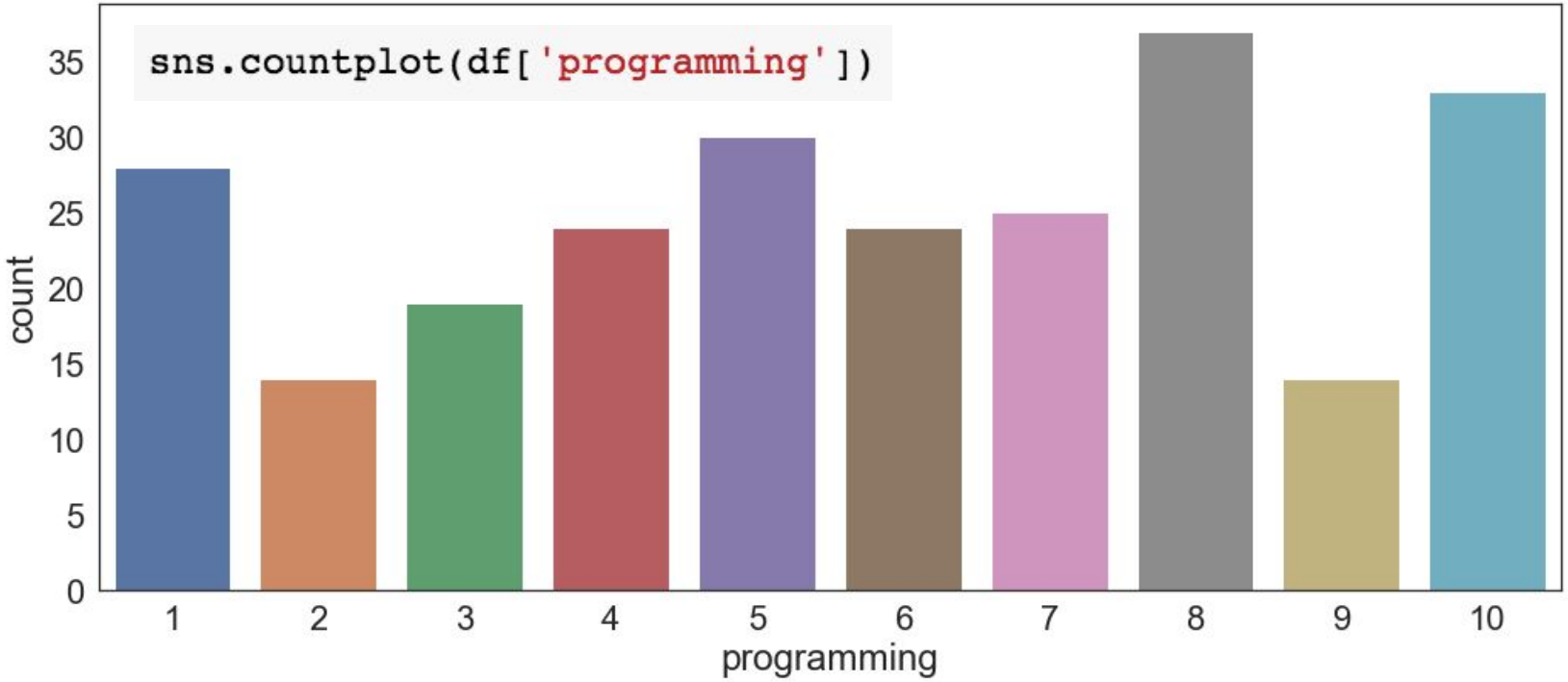
A Random Distribution



distribution has no apparent pattern

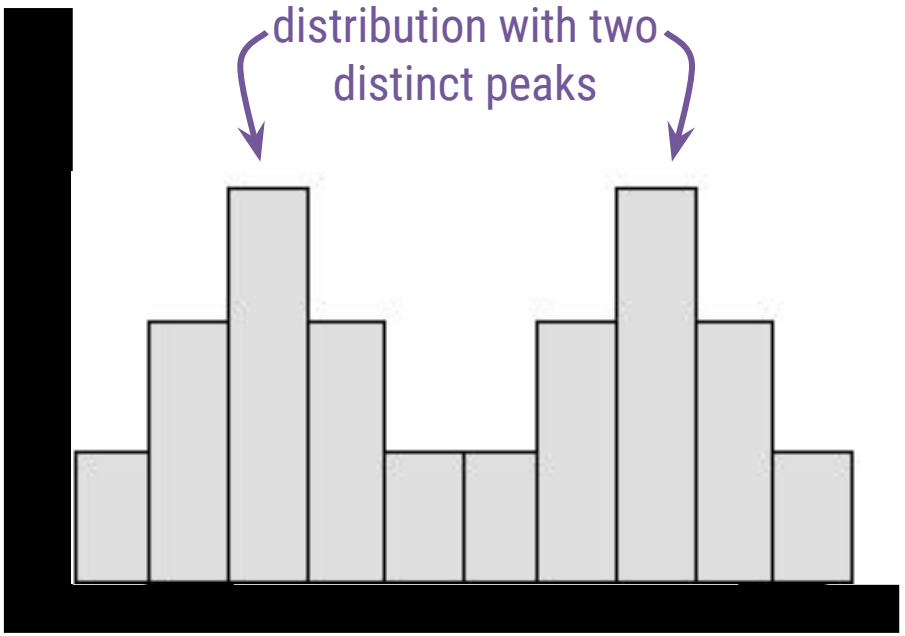
All distribution images adapted from: https://www.pqsystems.com/qualityadvisor/DataAnalysisTools/interpretation/histogram_shape.php

What about students' background experience in programming?



distribution has no apparent pattern; students' experience is all across the board

A Bimodal Distribution



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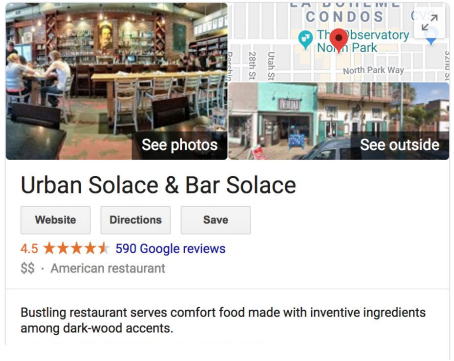
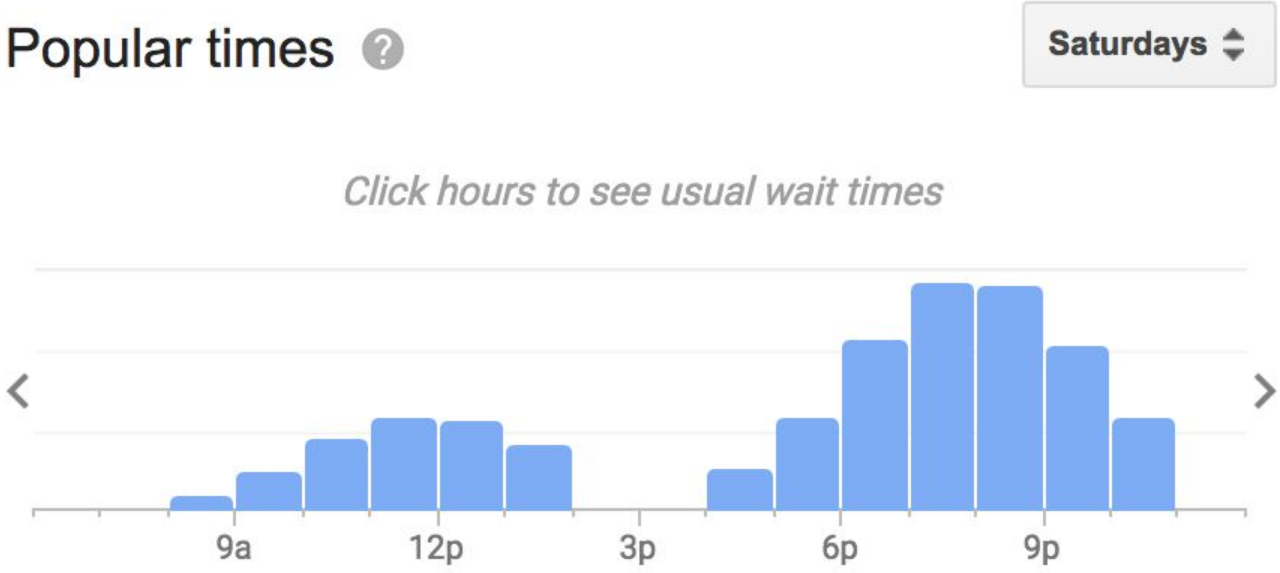
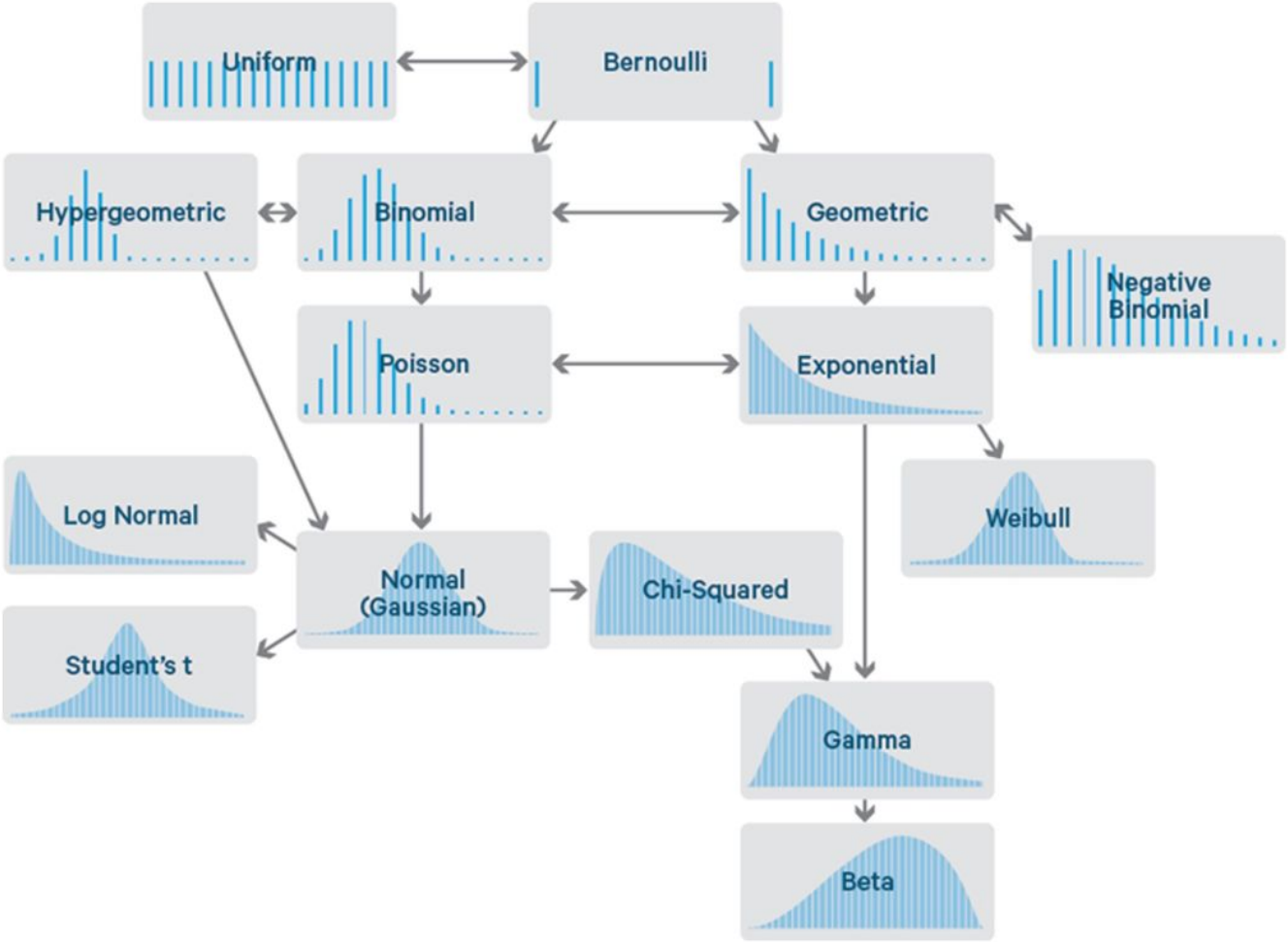
Popular times at a popular restaurant are often bimodal, with peaks here for brunch and dinner



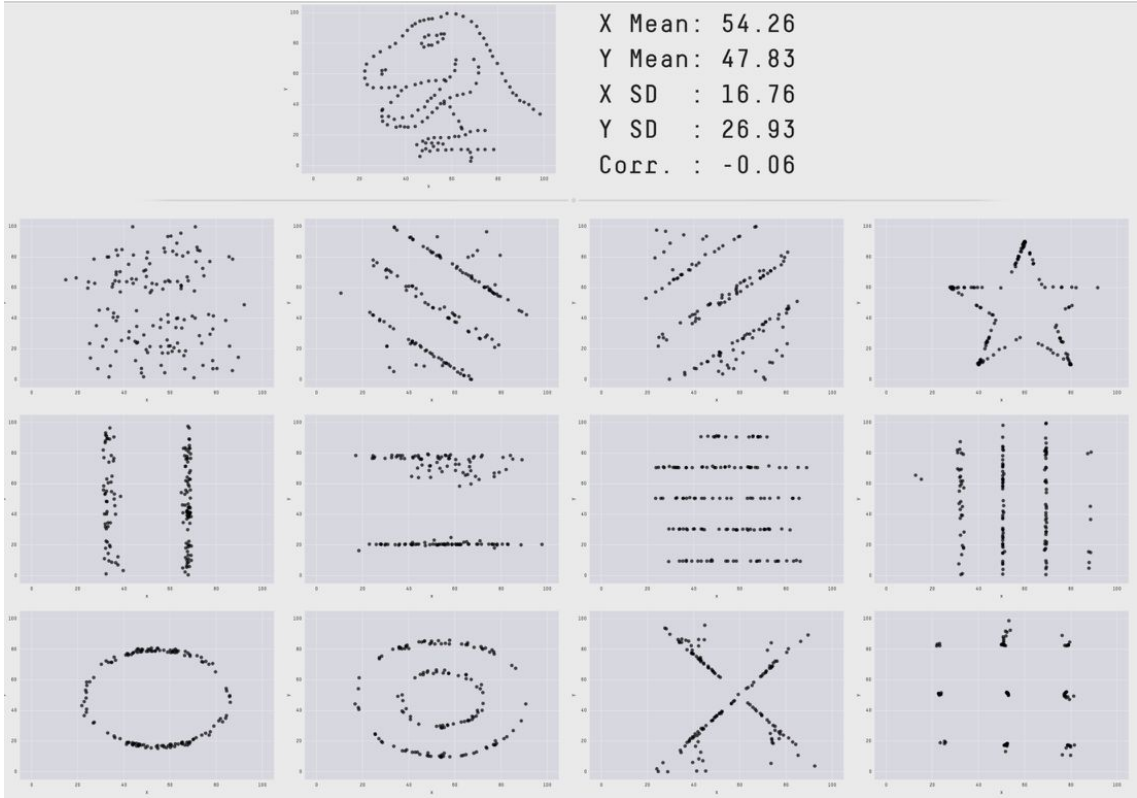


Source: https://www.google.com



Source: *Cloudera blog*: http://blog.cloudera.com/blog/2015/12/common-probability-distributions-the-data-scientists-crib-sheet/

Visualize your data



Source: Matejka & Fitzmaurice, *ACM SIGCHI* 2017

The shape of your data dictates what model you will use to answer your question.

2.3 Parsimony

Since all models are wrong the scientist cannot obtain a "correct" one by excessive elaboration. On the contrary following William of Occam he should seek an economical description of natural phenomena. Just as the ability to devise simple but evocative models is the signature of the great scientist so overelaboration and overparameterization is often the mark of mediocrity.

2.4 Worrying Selectively

Since all models are wrong the scientist must be alert to what is importantly wrong. It is inappropriate to be concerned about mice when there are tigers abroad.

CORRELATION

ASSOCIATION BETWEEN VARIABLES

i.e. Pearson Correlation, Spearman Correlation, chi-square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS BETWEEN VARIABLES

i.e. t-test, ANOVA

REGRESSION

DOES CHANGE IN ONE VARIABLE MEAN CHANGE IN ANOTHER?

I.e. simple regression, multiple regression

NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS IN THESE OTHER 3 CATEGORIES ARE NOT MET

i.e. Wilcoxon rank-sum test, Wilcoxon sign-rank test, sign test

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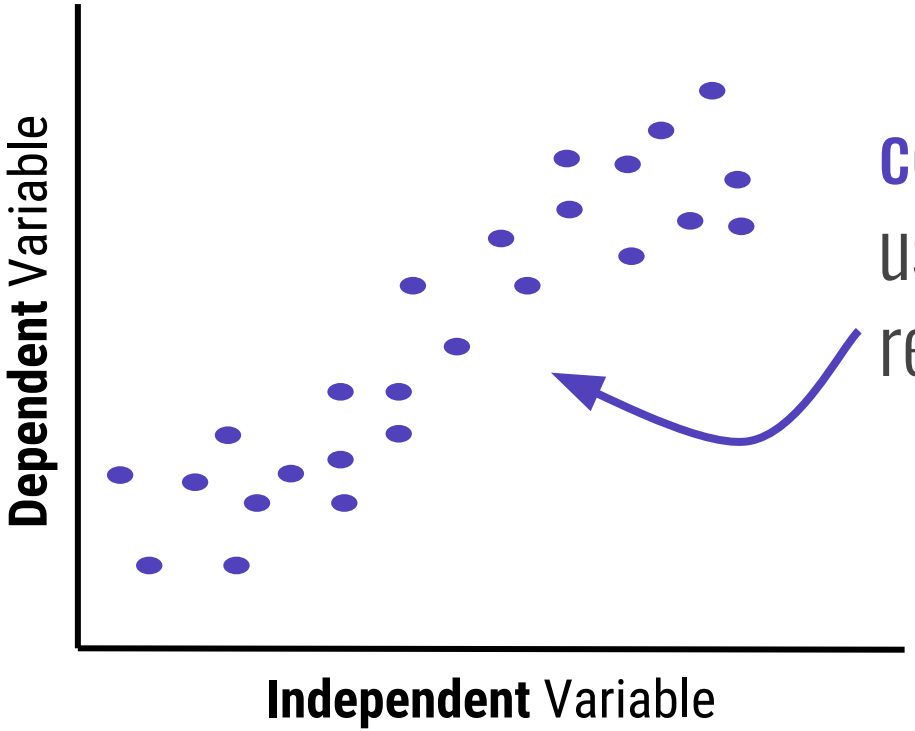
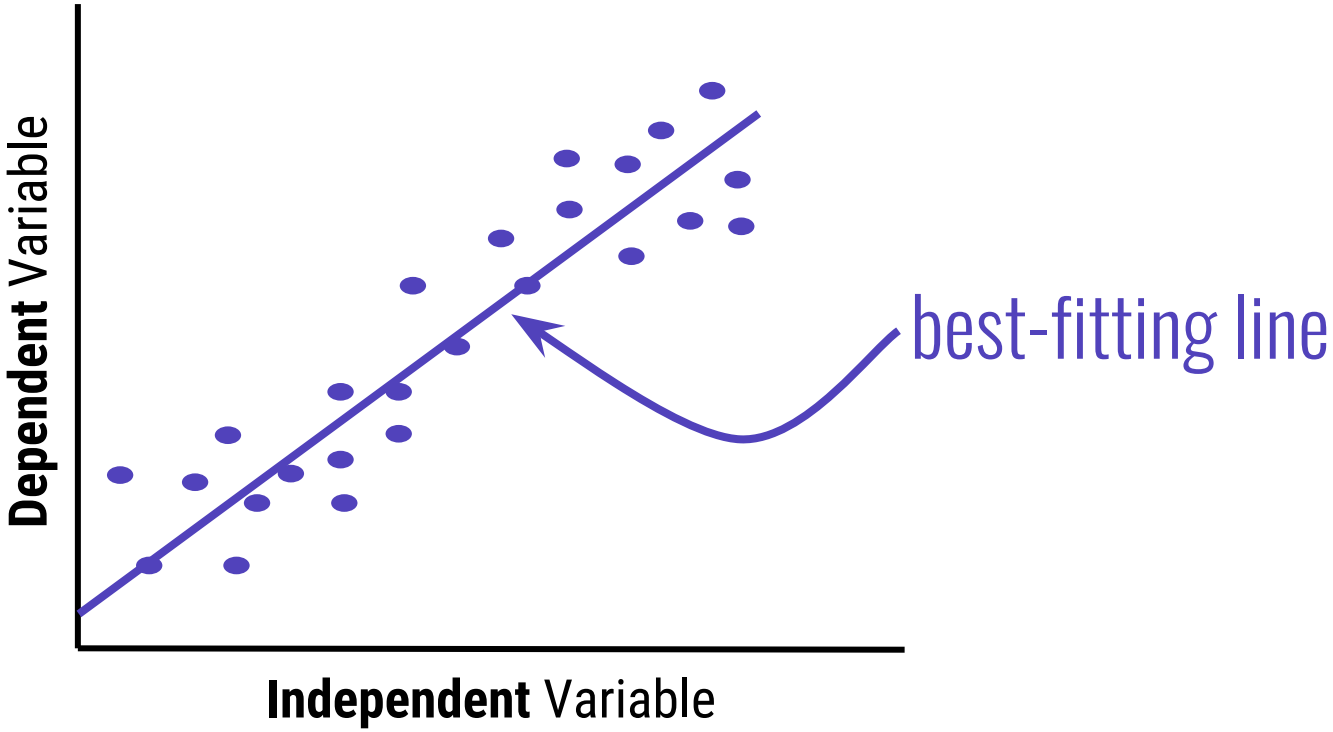
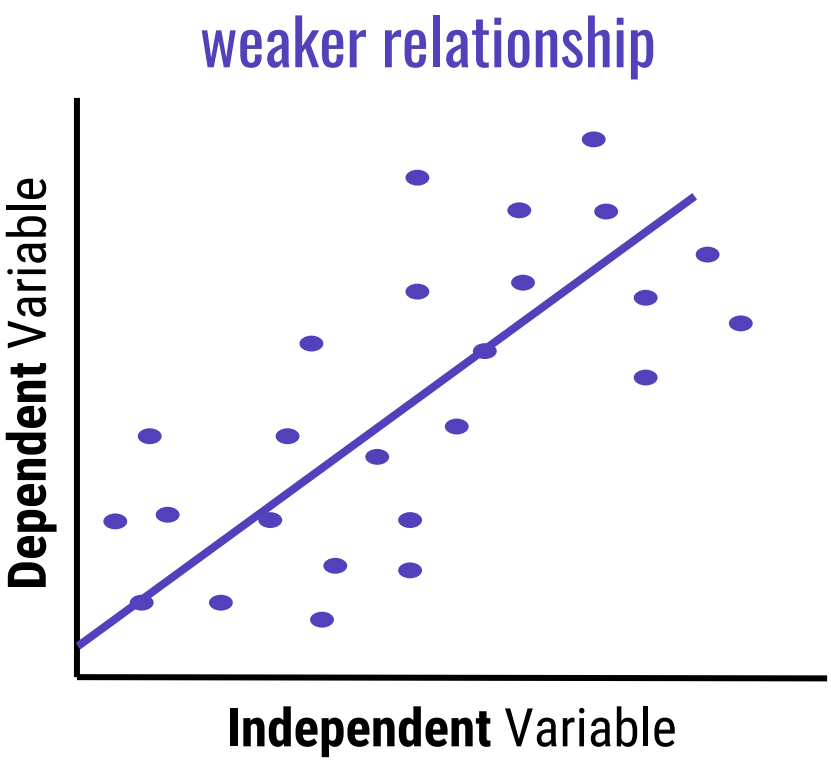
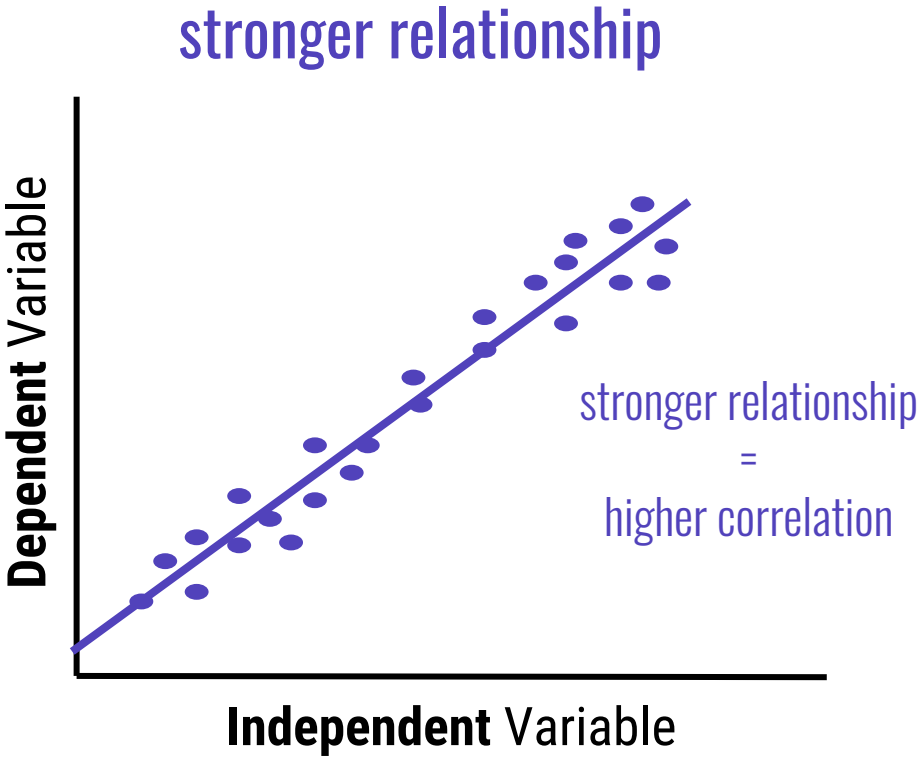


correlation can be used to describe this relationship


Dependent Variable
Independent Variable
best-fitting line

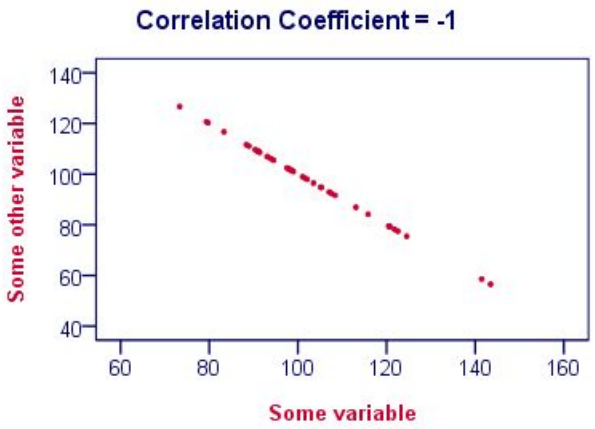

weaker relationship
Dependent Variable
Independent Variable

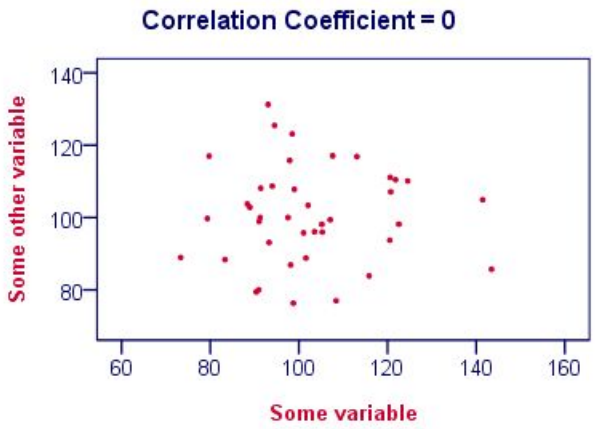

stronger relationship
Dependent Variable
Independent Variable
stronger relationship
=
higher correlation

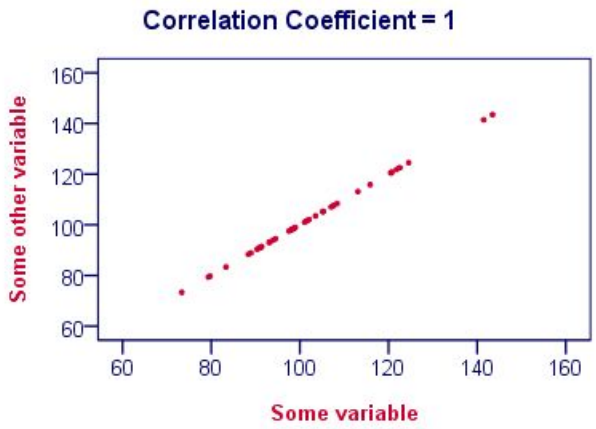
Pearson's *r* :

linear correlation between two variables

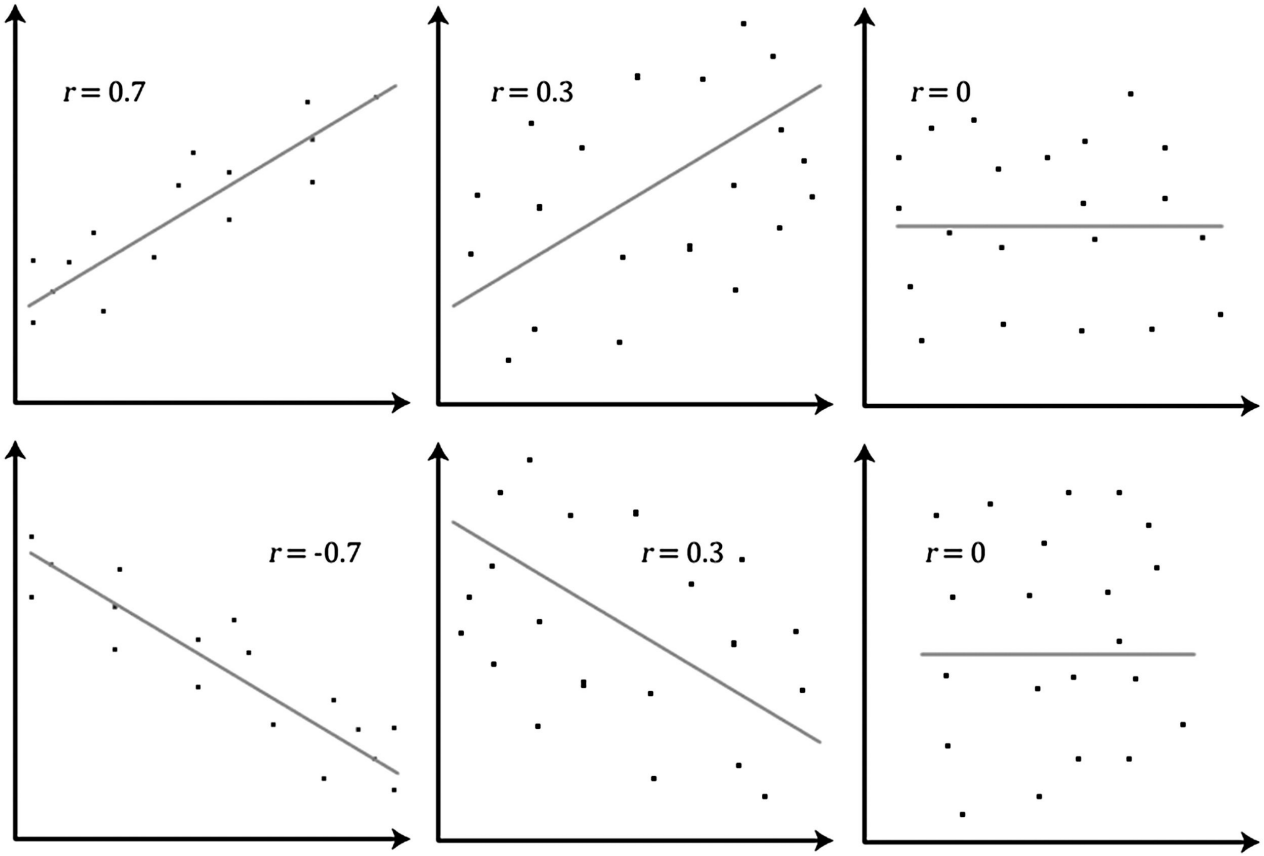
takes values [-1,1]



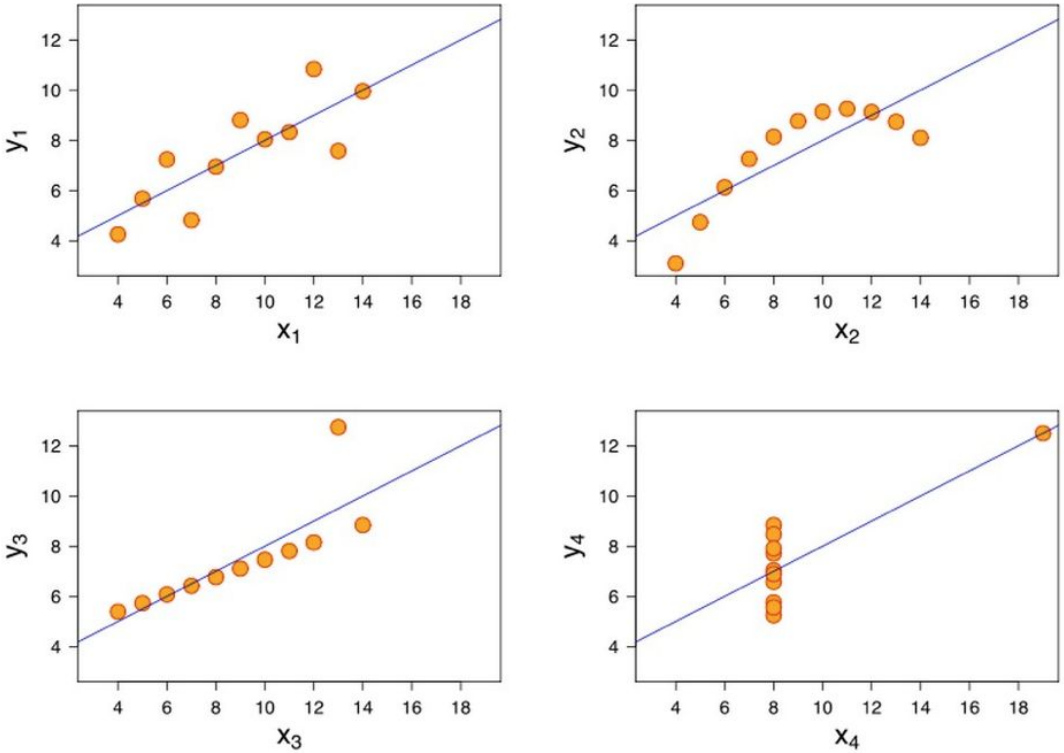




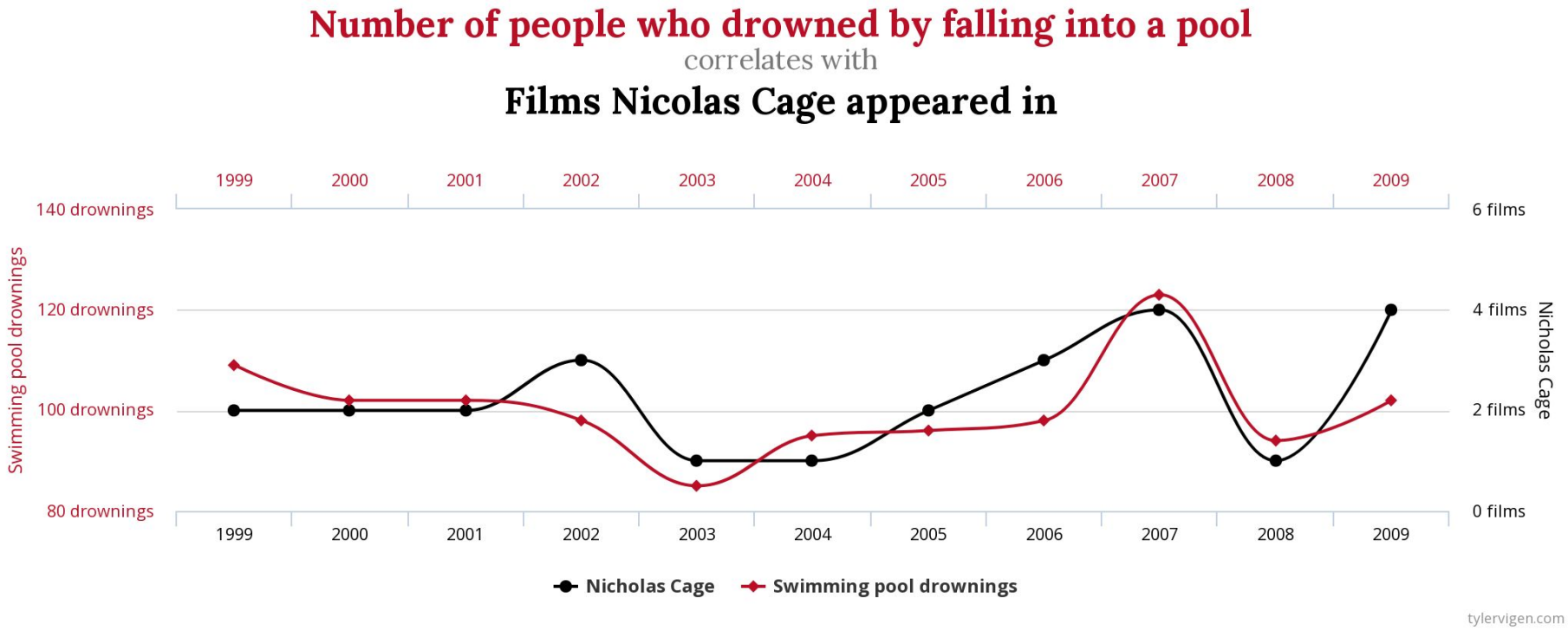
https://www.spss-tutorials.com/pearson-correlation-coefficient/


r = 0.7
r = 0.3
r = 0
r = -0.7
r = 0.3
r = 0

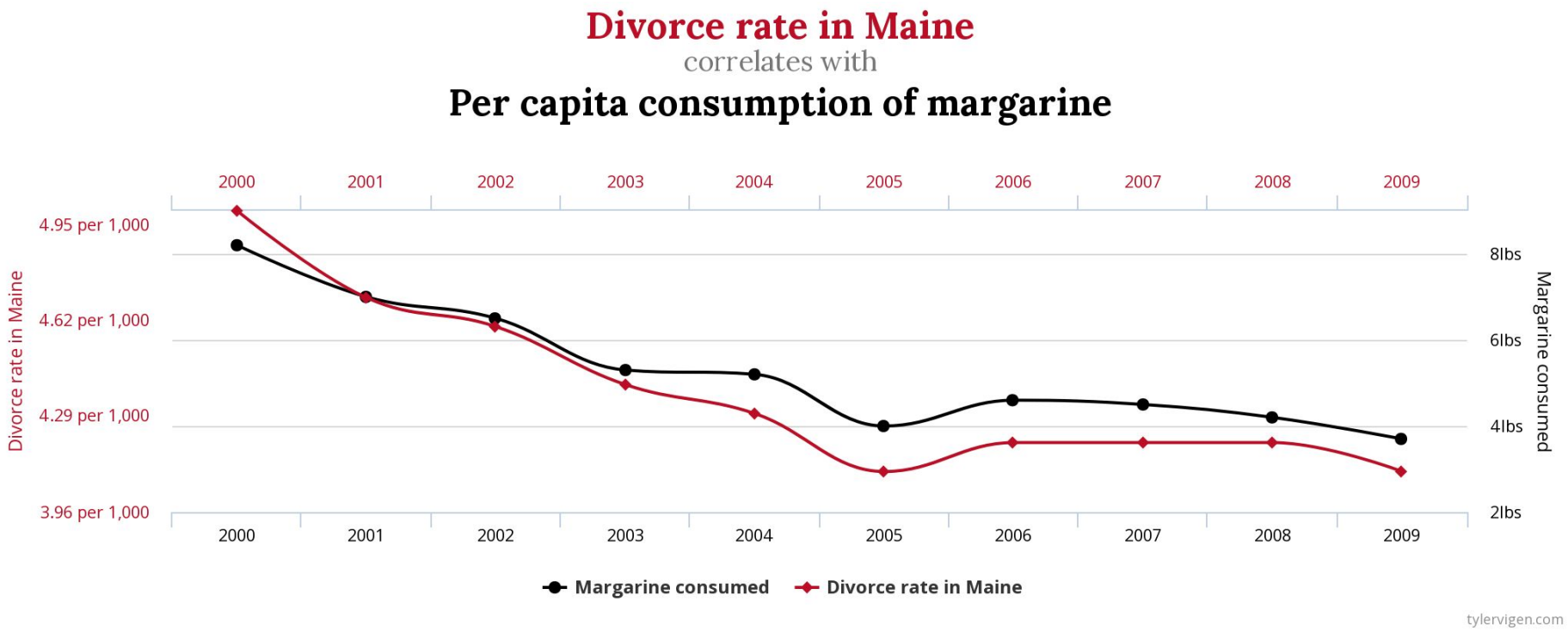
Property	Value
Mean of *x* in each case	9 (exact)
Variance of *x* in each case	11 (exact)
Mean of *y* in each case	7.50 (to 2 decimal places)
Variance of *y* in each case	4.122 or 4.127 (to 3 decimal places)
Correlation between *x* and *y* in each case	0.816 (to 3 decimal places)
Linear regression line in each case	$y = 3.00 + 0.500x$ (to 2 and 3 decimal places, respectively)



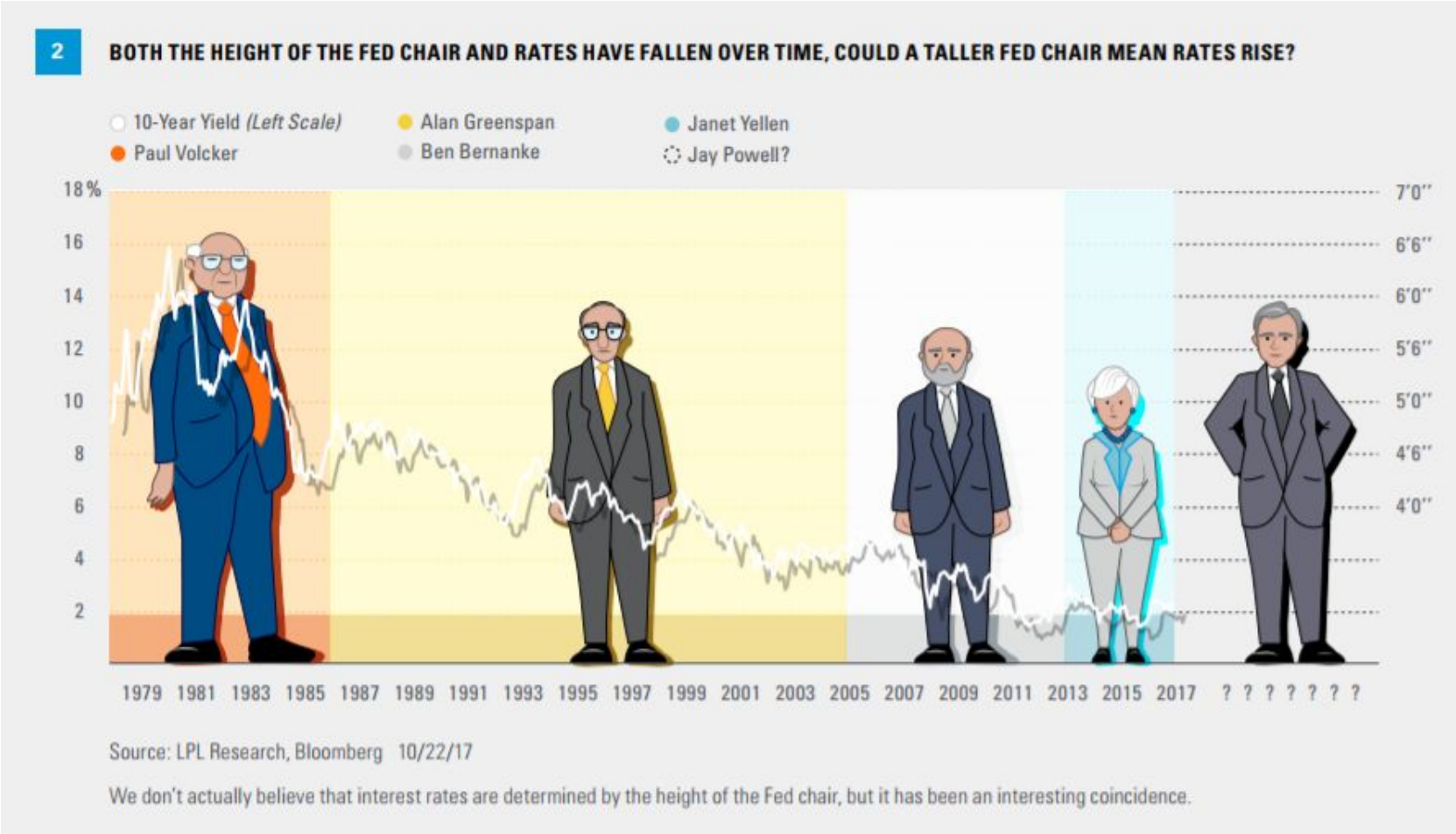
Anscombe's Quartet



http://www.tylervigen.com/spurious-correlations



http://www.tylervigen.com/spurious-correlations



https://www.cnbc.com/2017/10/26/height-matters-the-connection-between-tall-fed-chiefs-and-interest-rates.html

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i.e. t-test, ANOVA

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FOR WHEN ASSUMPTIONS IN THESE OTHER 3 CATEGORIES ARE NOT MET

i.e. Wilcoxon rank-sum test, Wilcoxon sign-rank test, sign test

Student’s t-test

Student's t-test

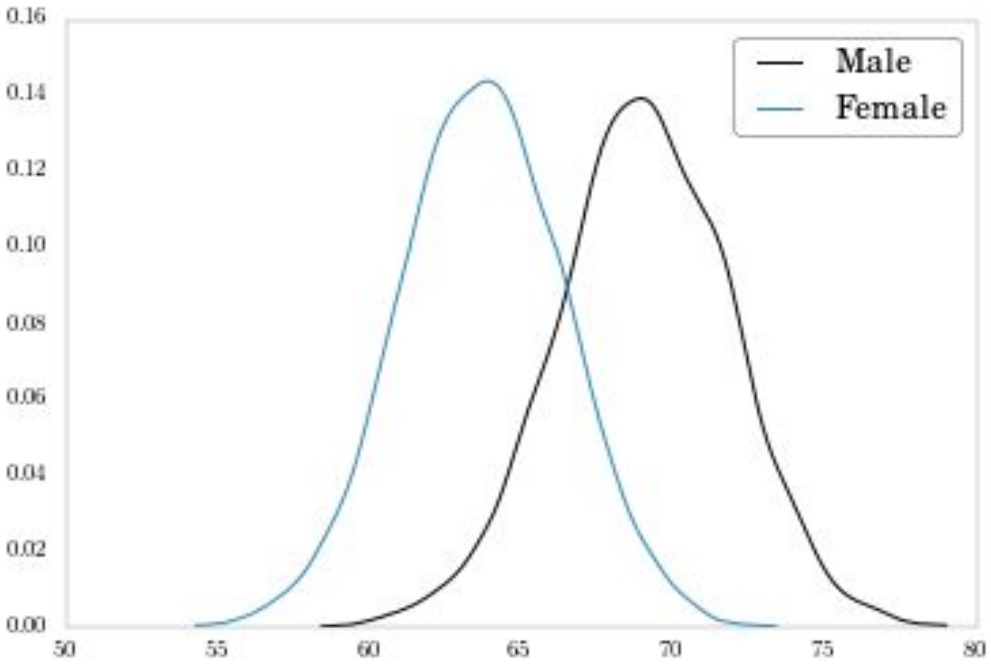
...brewing a batch of beer is a time consuming and expensive process, so in order to draw conclusions from experimental methods applied to just a few batches, Gosset had to figure out the role of chance in determining how a batch of beer had turned out. Guinness frowned upon academic publications, so Gosset had to publish his results under the modest pseudonym, "Student."

Source: Stanton, *An Introduction to Data Science*

t-test:

tests for difference in means between groups

Do the heights between males and females differ?



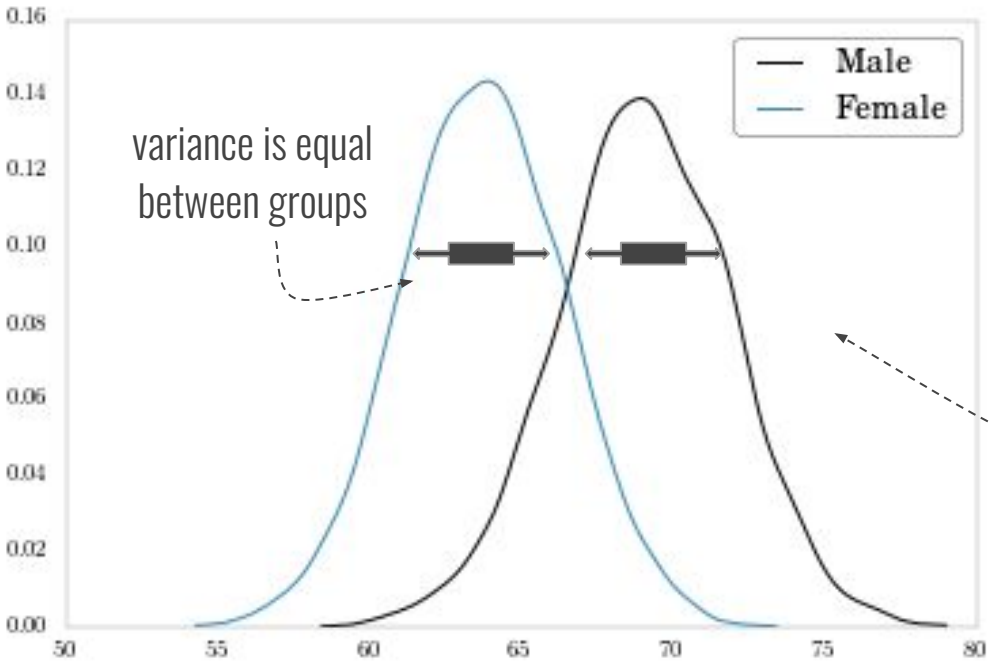
N=10,000

http://slendermeans.org/ml4h-ch2-p1.html

t-test Assumptions

1. Data are continuous
2. Normally distributed
3. Large enough sample size
4. Equal variance b/w groups

Do the heights between males and females differ?



variance is equal between groups

data are normally distributed

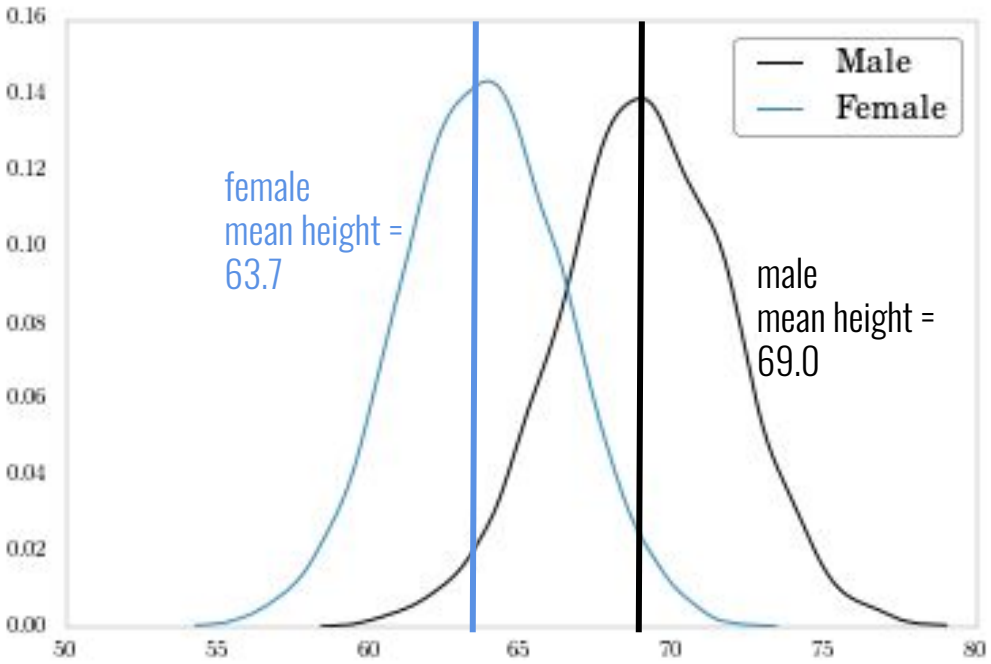
sample size is large enough

N=10,000

data are continuous

http://slendermeans.org/ml4h-ch2-p1.html

Do the heights between males and females differ?



N=10,000

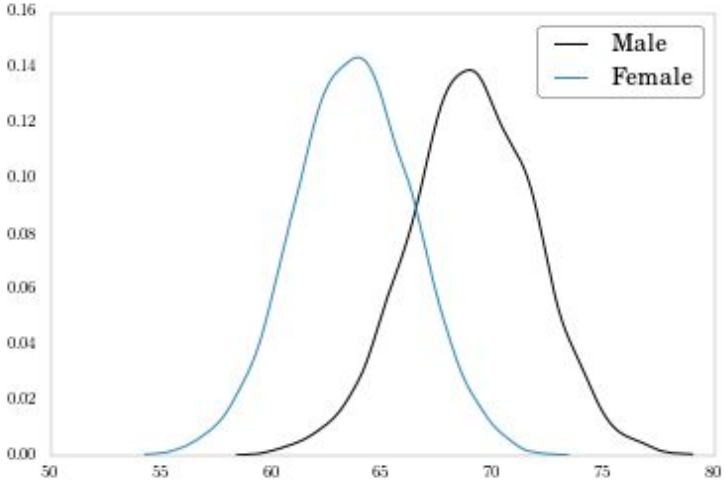
http://slendermeans.org/ml4h-ch2-p1.html

Do the heights between males and females differ?

t-statistic: -95.6

p-value << 0.001

95% CI for true difference in means [-5.43, -5.21]



CORRELATION

ASSOCIATION BETWEEN VARIABLES

i.e. Pearson Correlation, Spearman Correlation, chi-square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS BETWEEN VARIABLES

i.e. t-test, ANOVA

REGRESSION

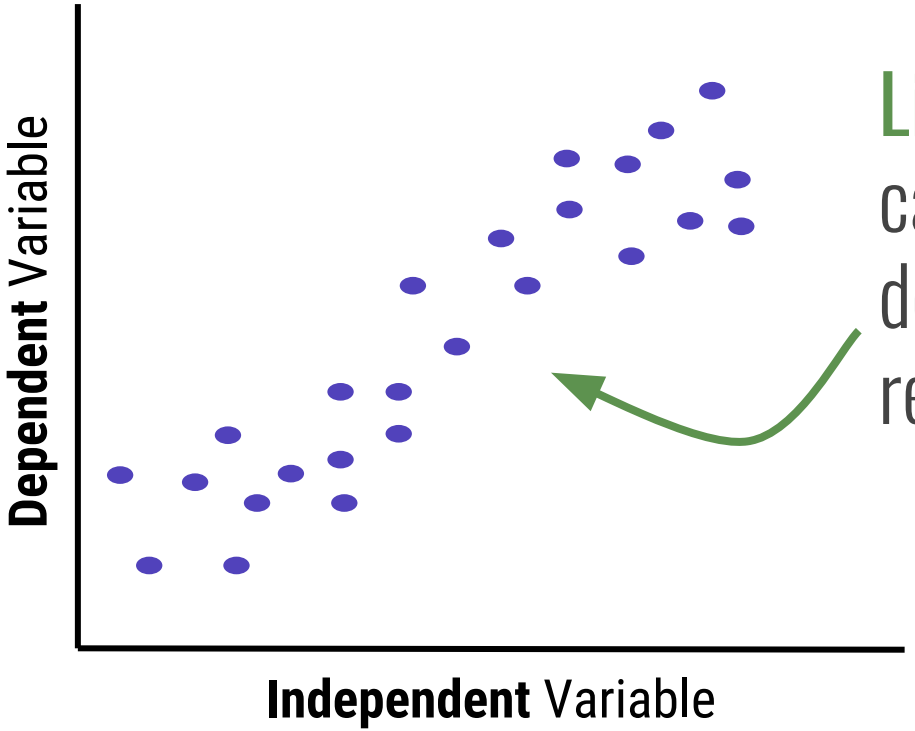
DOES CHANGE IN ONE VARIABLE MEAN CHANGE IN ANOTHER?

I.e. simple regression, multiple regression

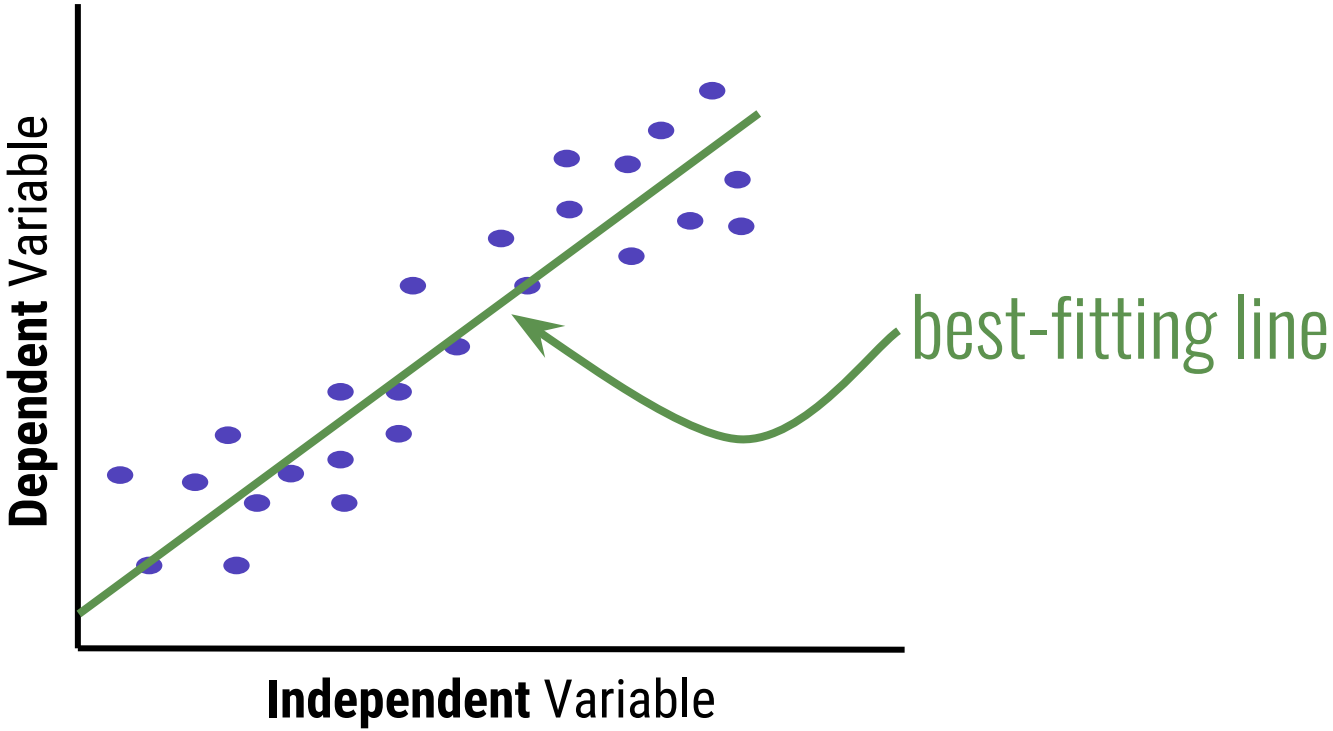
NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS IN THESE OTHER 3 CATEGORIES ARE NOT MET

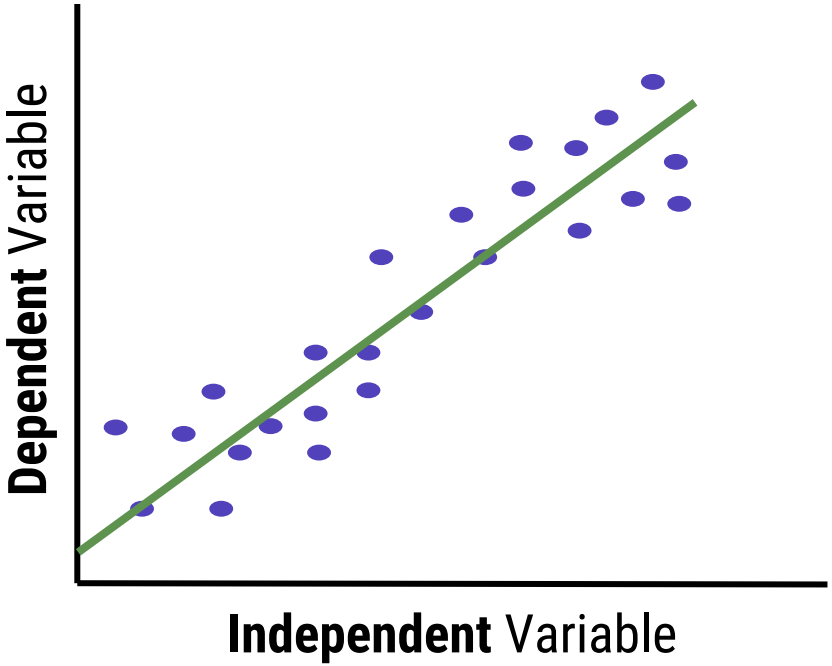
i.e. Wilcoxon rank-sum test, Wilcoxon sign-rank test, sign test



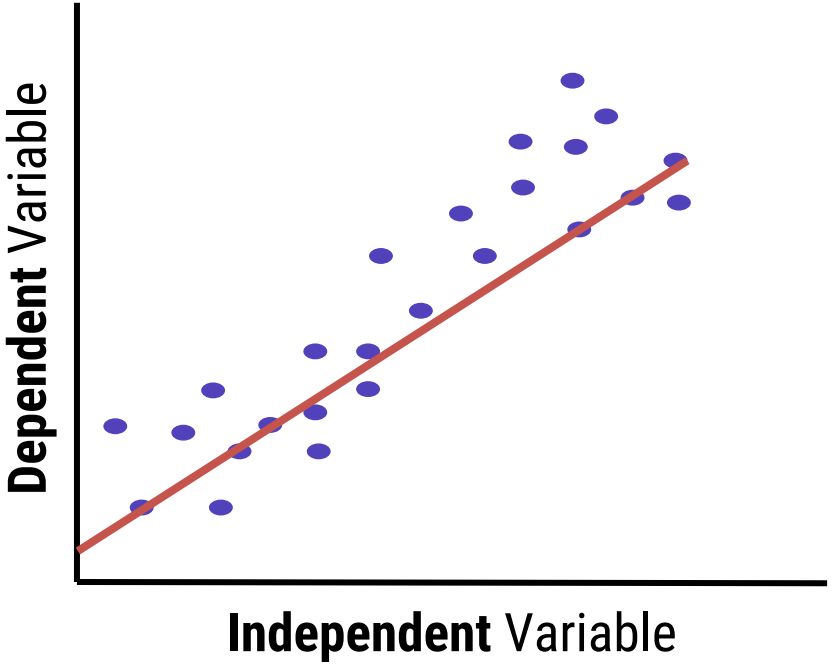
Linear regression can be used to describe this relationship

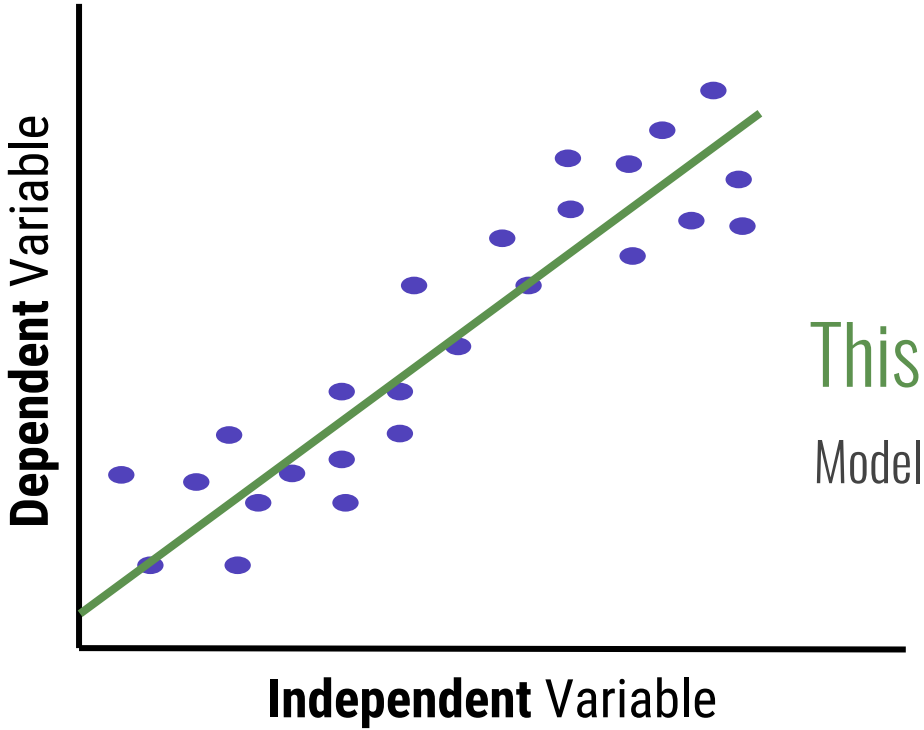

Dependent Variable
Independent Variable
best-fitting line

Best-fitting line



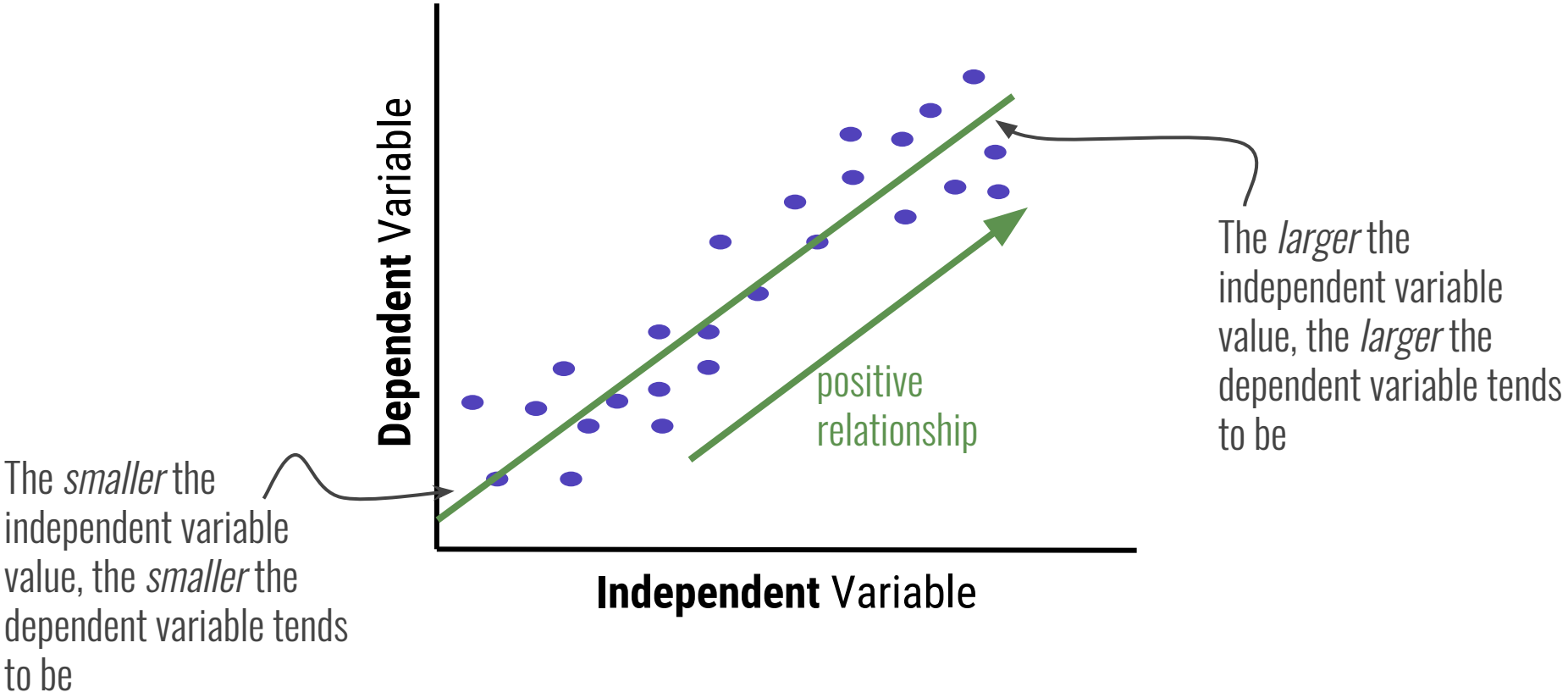
NOT a best-fitting line

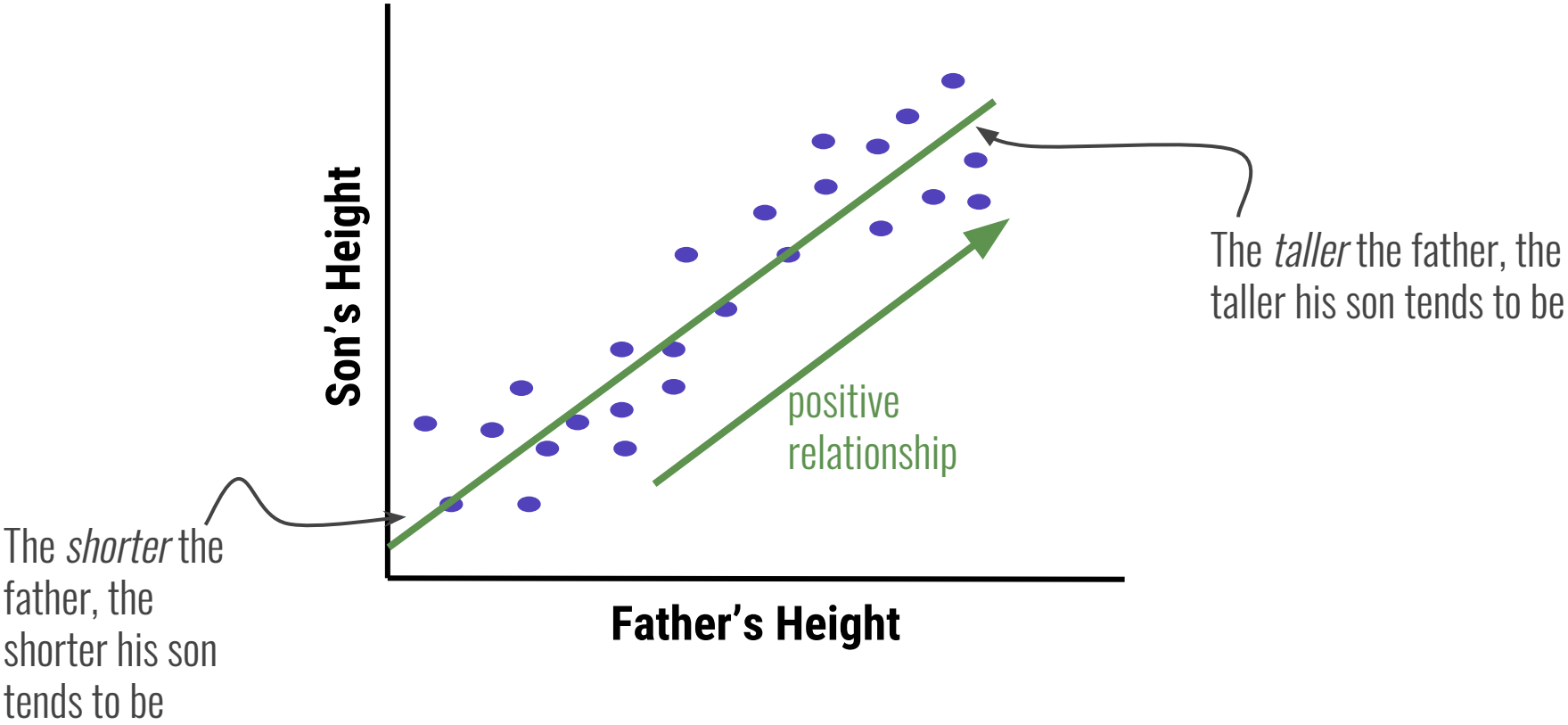


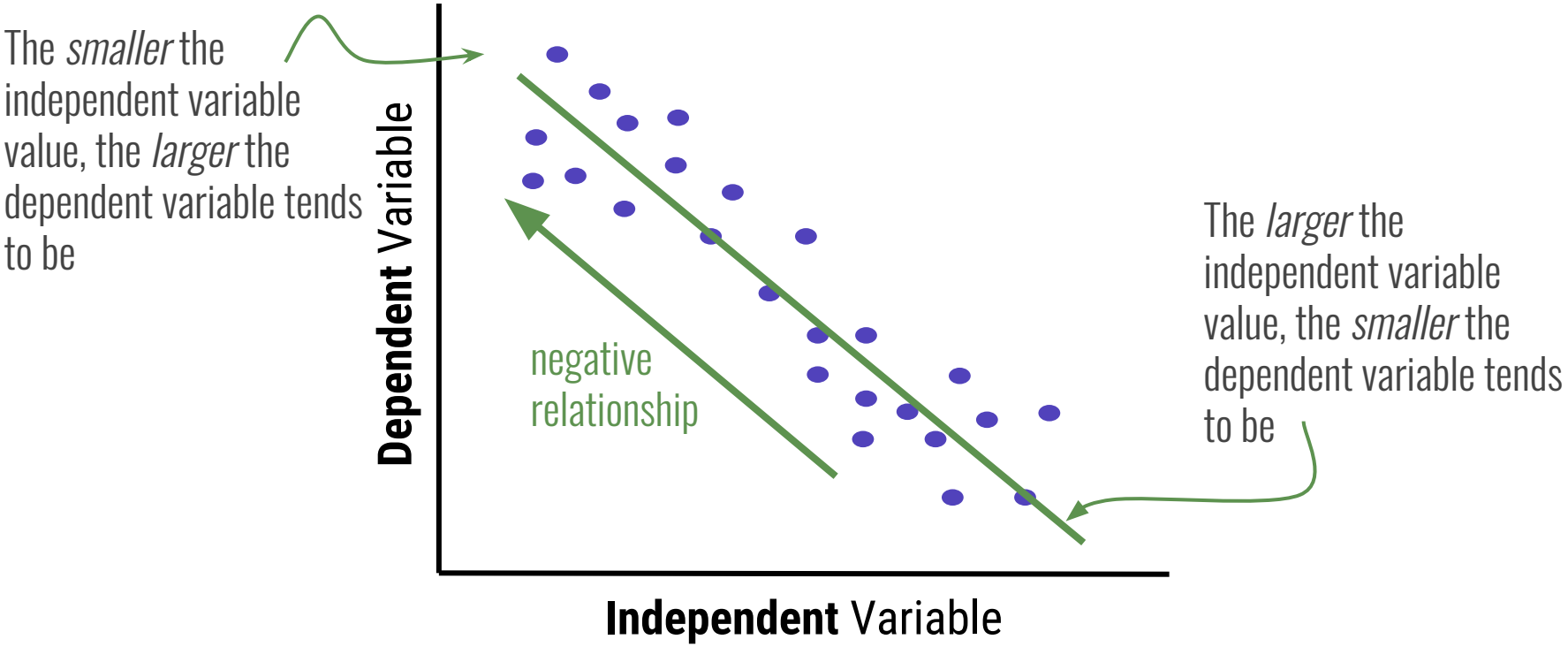


This line is a **model** of the data

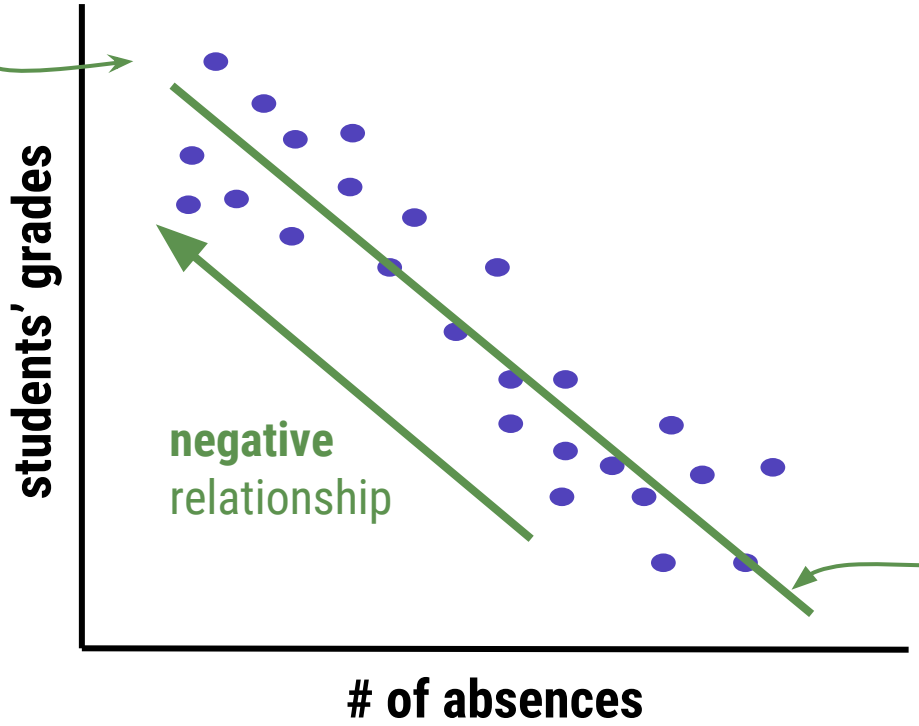
Models are mathematical equations generated to *represent* the real life situation


Dependent Variable
Independent Variable
positive relationship
The *larger* the independent variable value, the *larger* the dependent variable tends to be
The *smaller* the independent variable value, the *smaller* the dependent variable tends to be


Son's Height
Father's Height
positive relationship
The *taller* the father, the taller his son tends to be
The *shorter* the father, the shorter his son tends to be

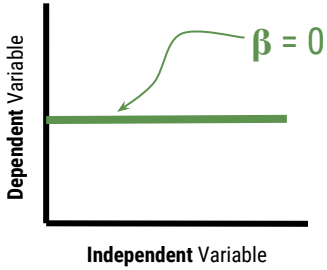

The *smaller* the independent variable value, the *larger* the dependent variable tends to be
Dependent Variable
negative relationship
The *larger* the independent variable value, the *smaller* the dependent variable tends to be
Independent Variable

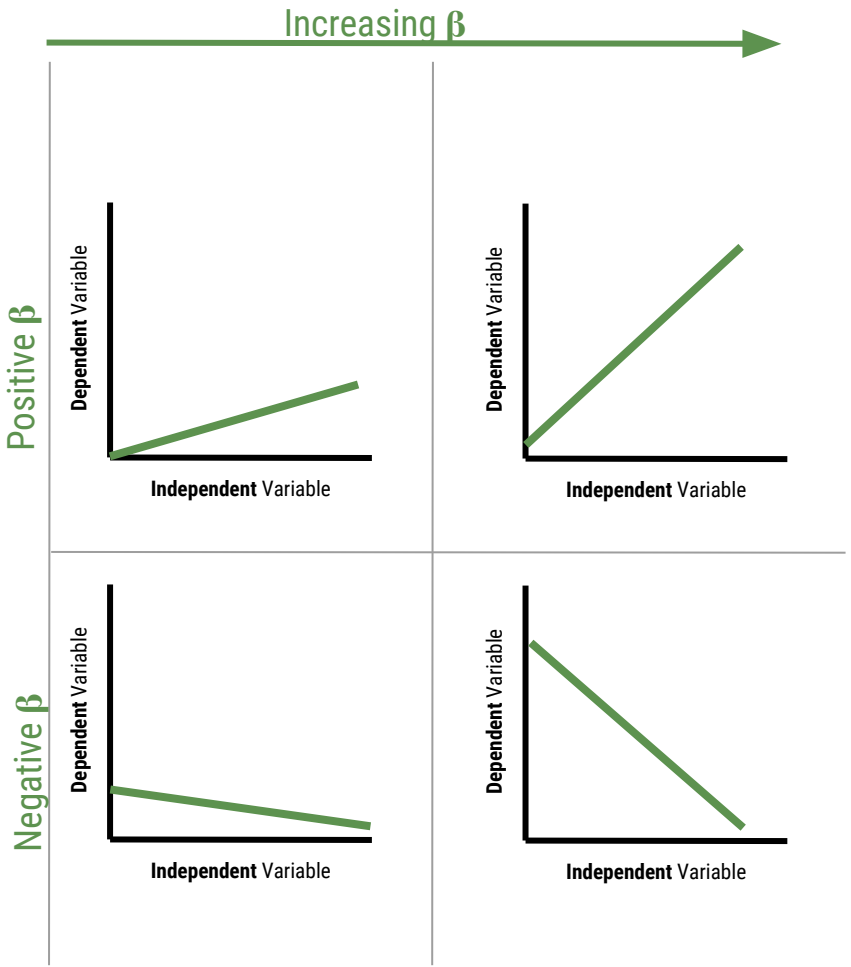
The *lower* the number of absences, the *higher* the students' grades tend to be

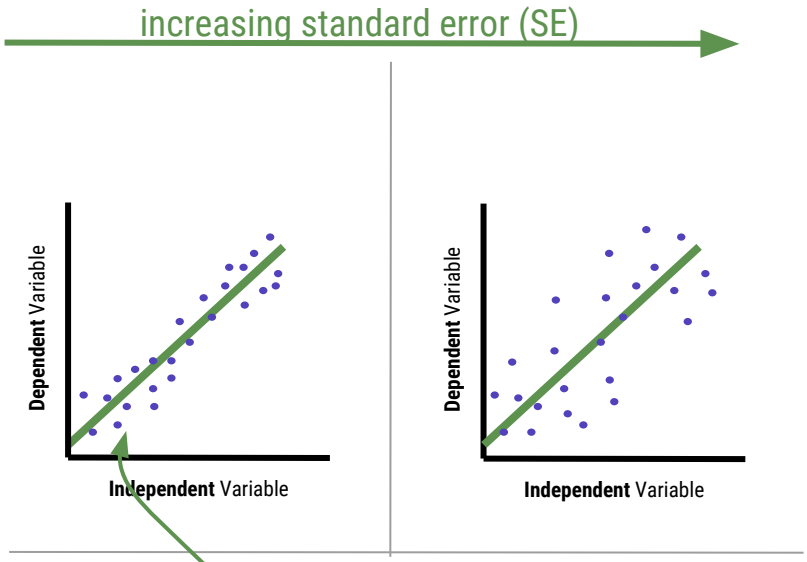


The more absences, the lower the students' grades

Effect size can be estimated using the slope of the line







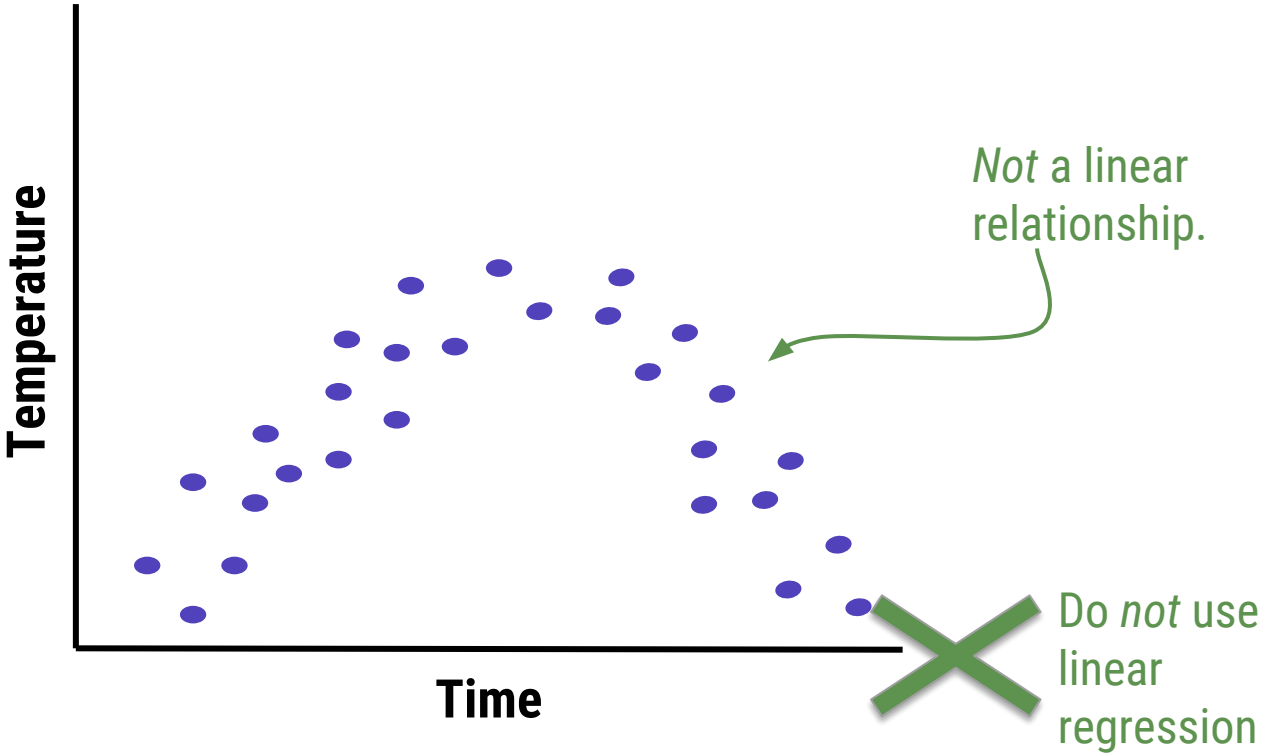
The *closer* the points are to the regression line, the *more uncertain* we are in our estimate

Takes into account the effect size (β) and the SE

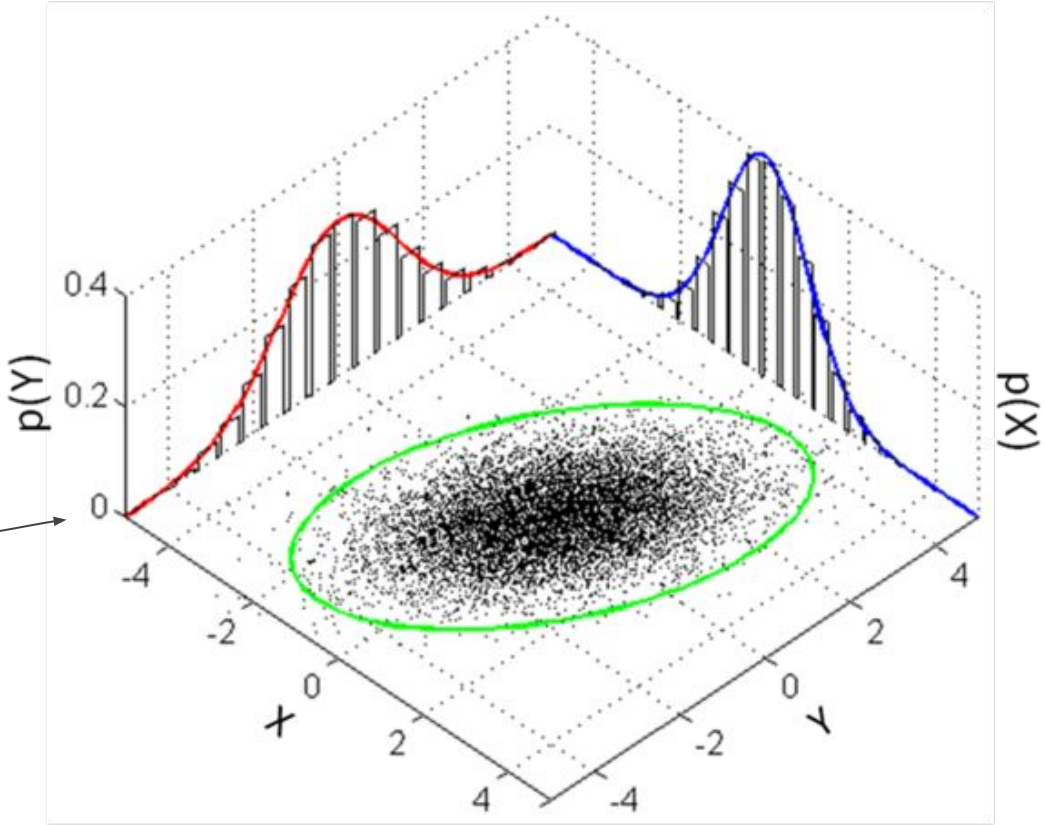
p-value : the probability of getting the observed results (or results more extreme) by chance alone

Assumptions of linear regression

1. Linear relationship
2. Multivariate normality
3. No multicollinearity
4. No auto-correlation
5. Homoscedasticity

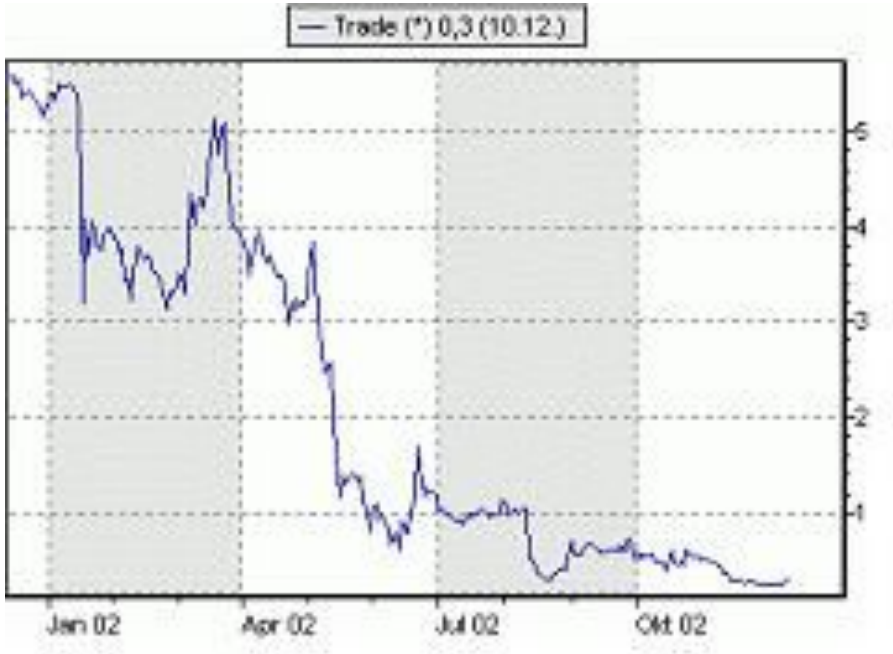

Temperature
Time
Not a linear relationship.
Do *not* use linear regression

A **multivariate normal** probability distribution (joint normal)

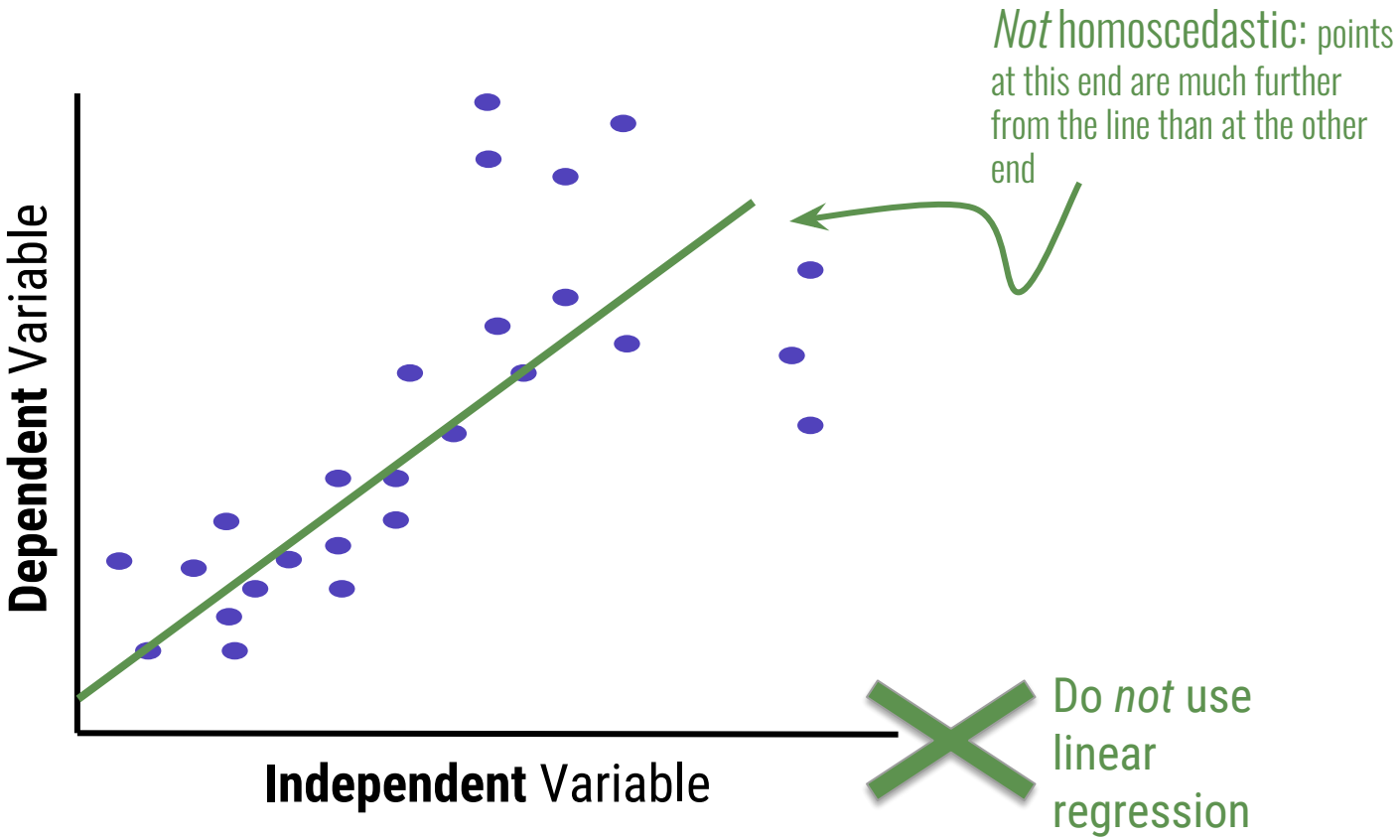


Linear regression assumes no multicollinearity. **Multicollinearity** occurs when the independent variables are too highly correlated with each other.

Autocorrelation occurs when the observations are *not* independent of one another (i.e. stock prices)



https://www.statisticssolutions.com/assumptions-of-linear-regression/

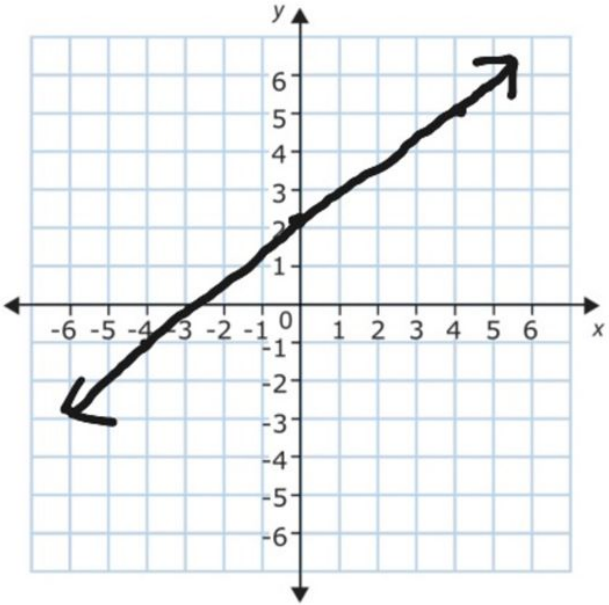

Not homoscedastic: points at this end are much further from the line than at the other end
Dependent Variable
Independent Variable
Do *not* use linear regression

A line

slope

$$y = mx + b$$

intercept



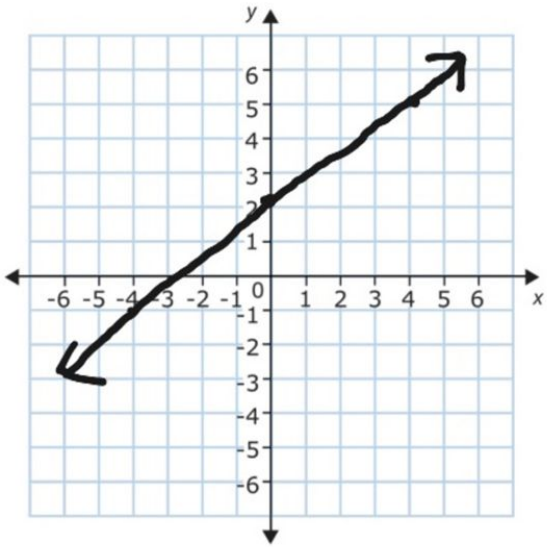
$$y = \frac{3}{4}x + 2$$

Ordinary Least Squares (OLS)

slope

$$y = mx + b$$

intercept



$$y = \frac{3}{4}x + 2$$

intercept

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

Ordinary Least Squares (OLS)

$$y_i = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} x_i + \overbrace{\varepsilon_i}^{\text{error}}$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

Goal: find values for the parameters to provide a "best" fit for the data

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

Goal: find values for the parameters to provide a "best" fit for the data

....how?

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

minimize this

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

minimize this

...how?

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

$$\min \sum_{i=1}^{n} \varepsilon_i^2$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

...why squared?

$$\min \sum_{i=1}^{n} \varepsilon_i^2$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

...why squared?

$$\min \sum_{i=1}^{n} \varepsilon_i^2$$

$$y_i - \beta_0 - \beta_1 x_i = \varepsilon_i$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

...why squared?

$$\min \sum_{i=1}^{n} \varepsilon_i^2$$

$$y_i - \beta_0 - \beta_1 x_i = \varepsilon_i$$

$$\min \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2)$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

$$\beta_1 = r_{xy} \frac{\theta_y}{\theta_x}$$

$$\beta_0 = \mu_y - \beta_1 \mu_x$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

$$\beta_1 = \rho \frac{\theta_y}{}$$

What does that even mean?

$$\beta_0 = \mu_y - \rho_1 \mu_x$$

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

$$\beta_1 = r_{xy} \frac{\theta_y}{\theta_x}$$

when the input increases by σ_x, the prediction increases by the x,y correlation

Ordinary Least Squares (OLS)

parameter

intercept

residual

error

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

slope

parameter

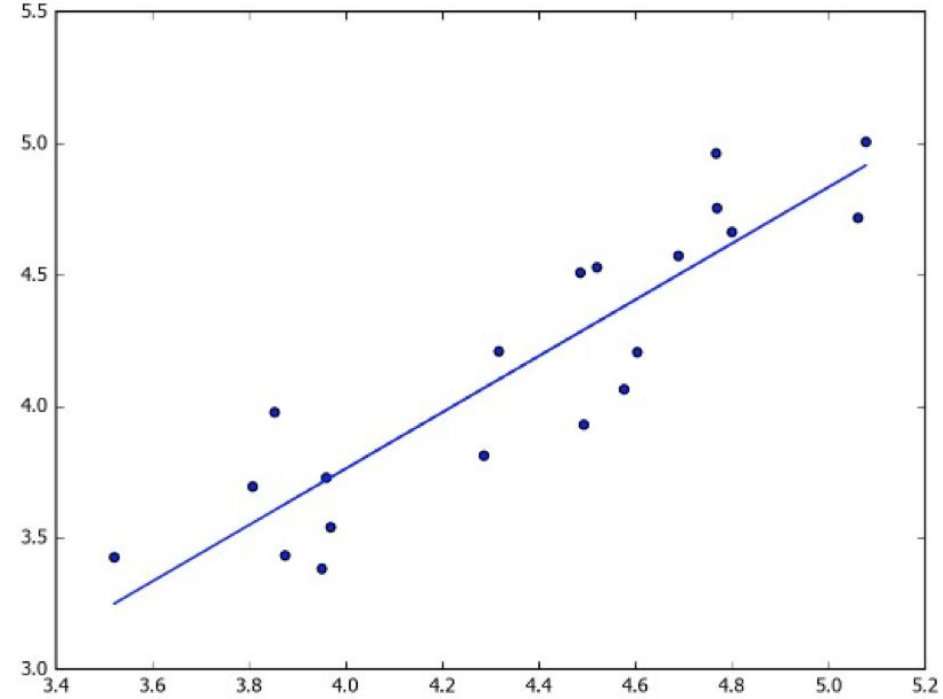
$$\beta_1 = r_{xy} \frac{\theta_y}{\theta_x}$$

when the input increases by σ_x, the prediction increases by the *x,y* correlation

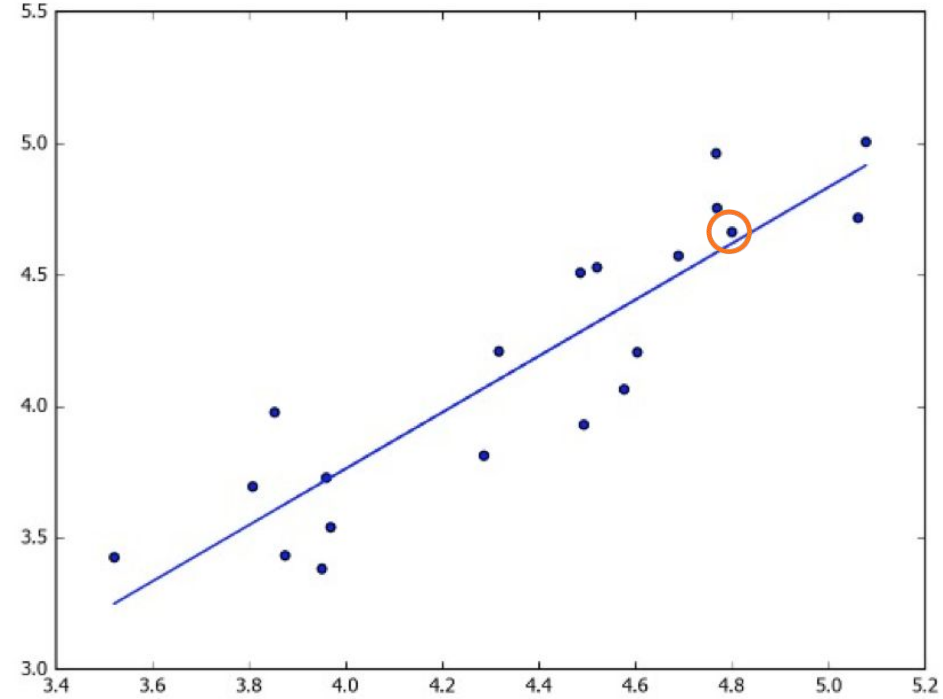
$$\beta_0 = \mu_y - \beta_1 \mu_x$$

when we see the average value of x, we predict the average value of y

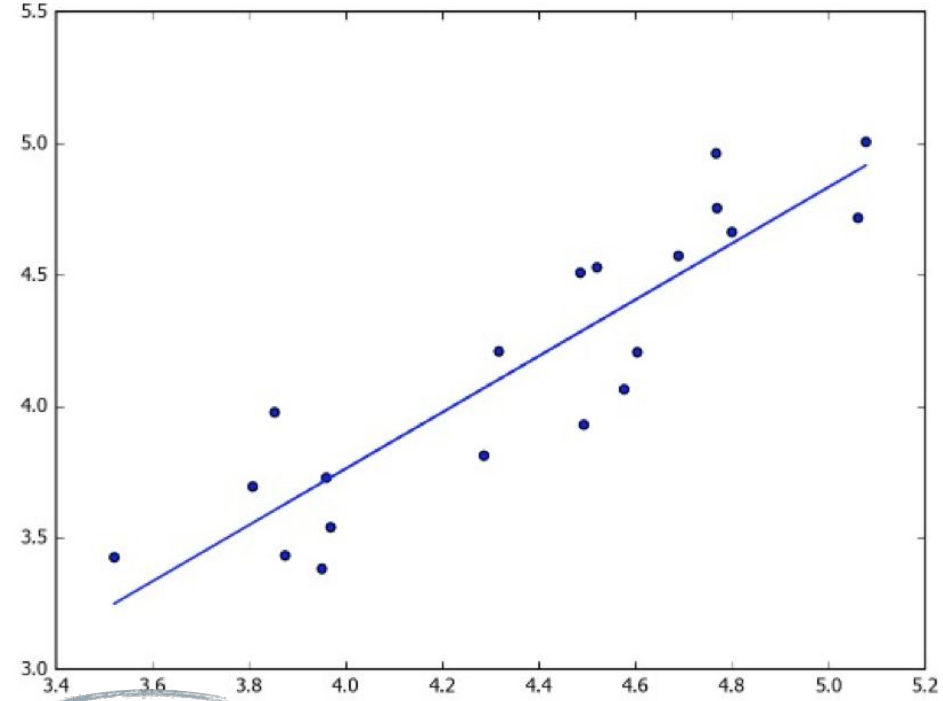
Outliers



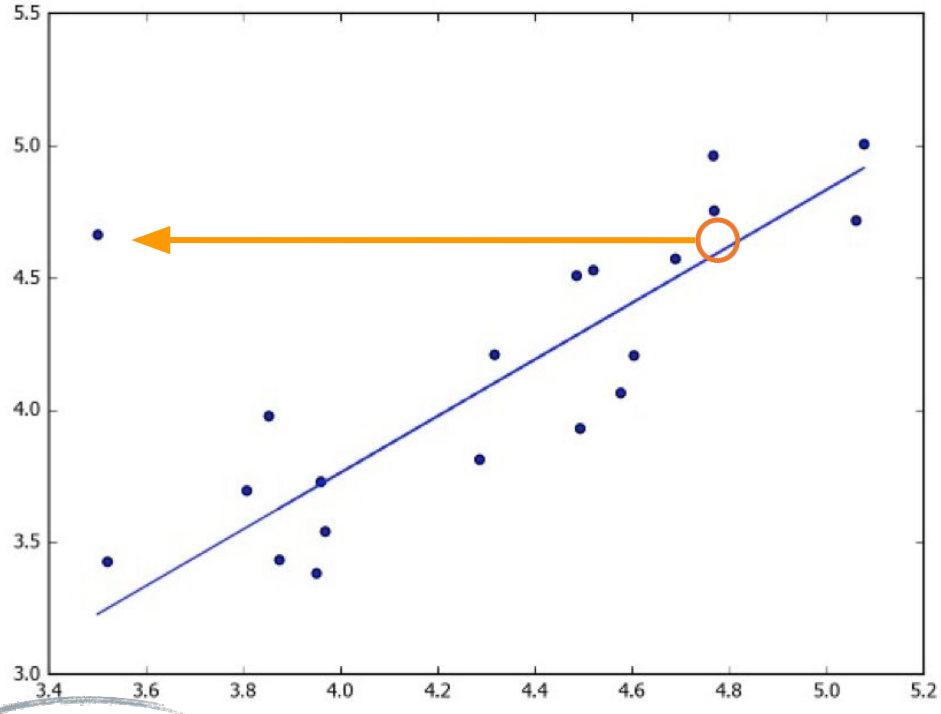

```
r = 0.899192316859 ; p = 7.04649901141e-08
```



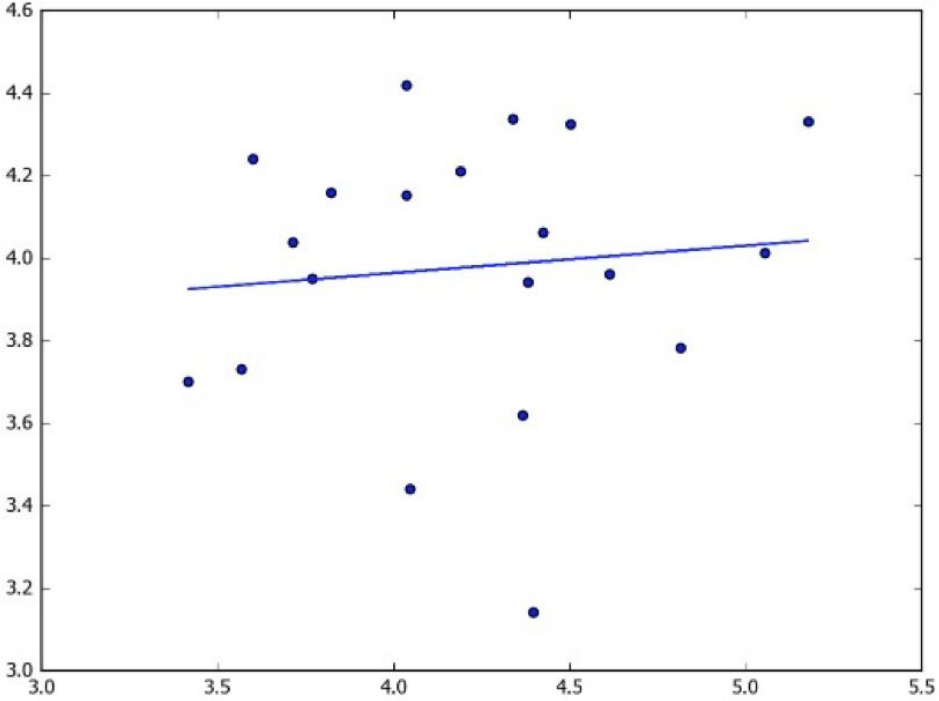

```
r = 0.899192316859 ; p = 7.04649901141e-08
```



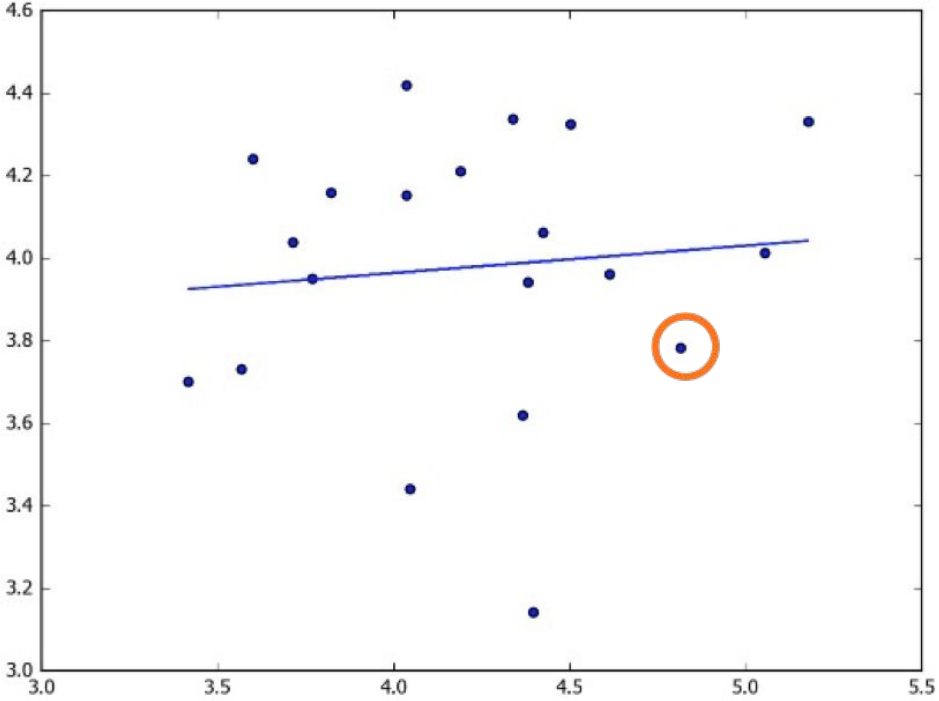
r = 0.899192316859 ; p = 7.04649901141e-08



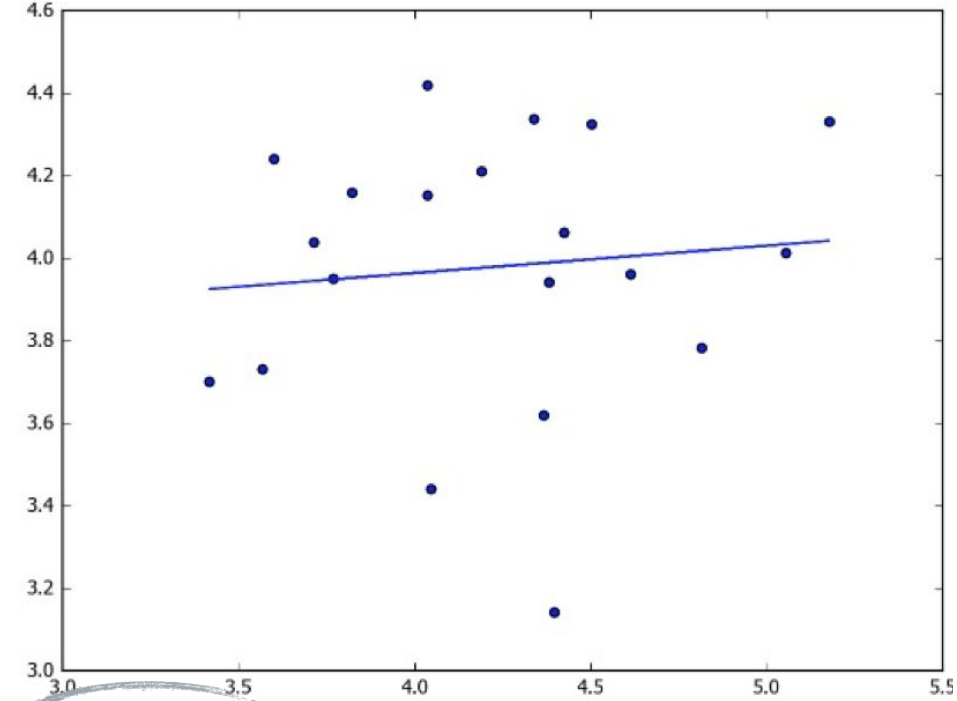
0.709892176011 0.000454396182138



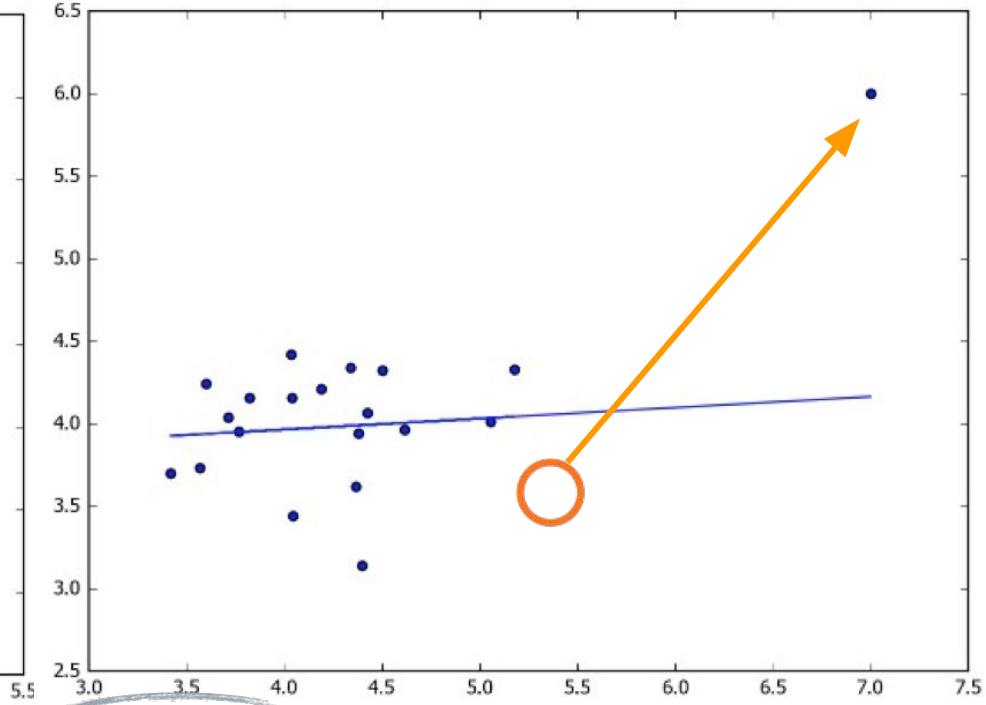

```
r = 0.0982029696446 ; p = 0.680415082529
```




```
r = 0.0982029696446 ; p = 0.680415082529
```

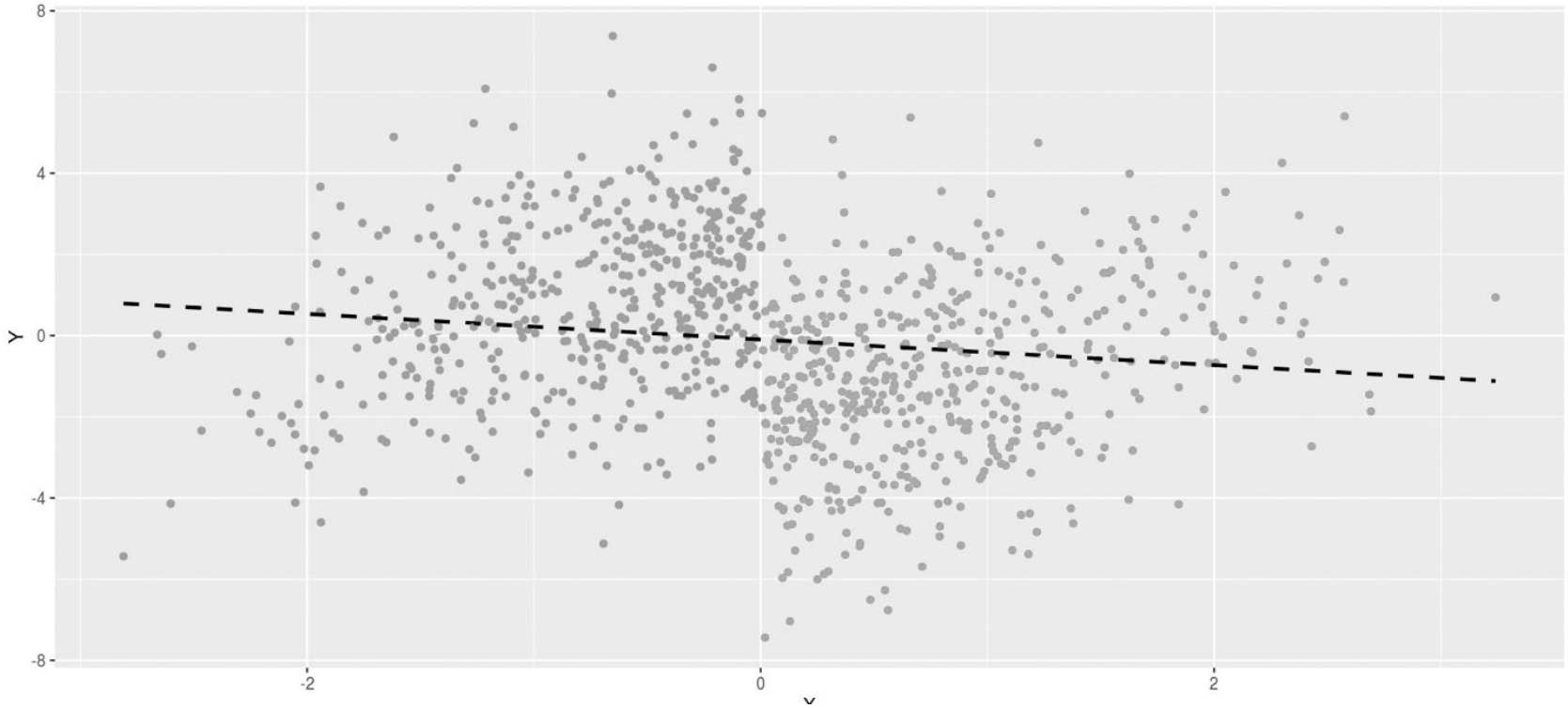


r = 0.098202969644 ; p = 0.680415082529



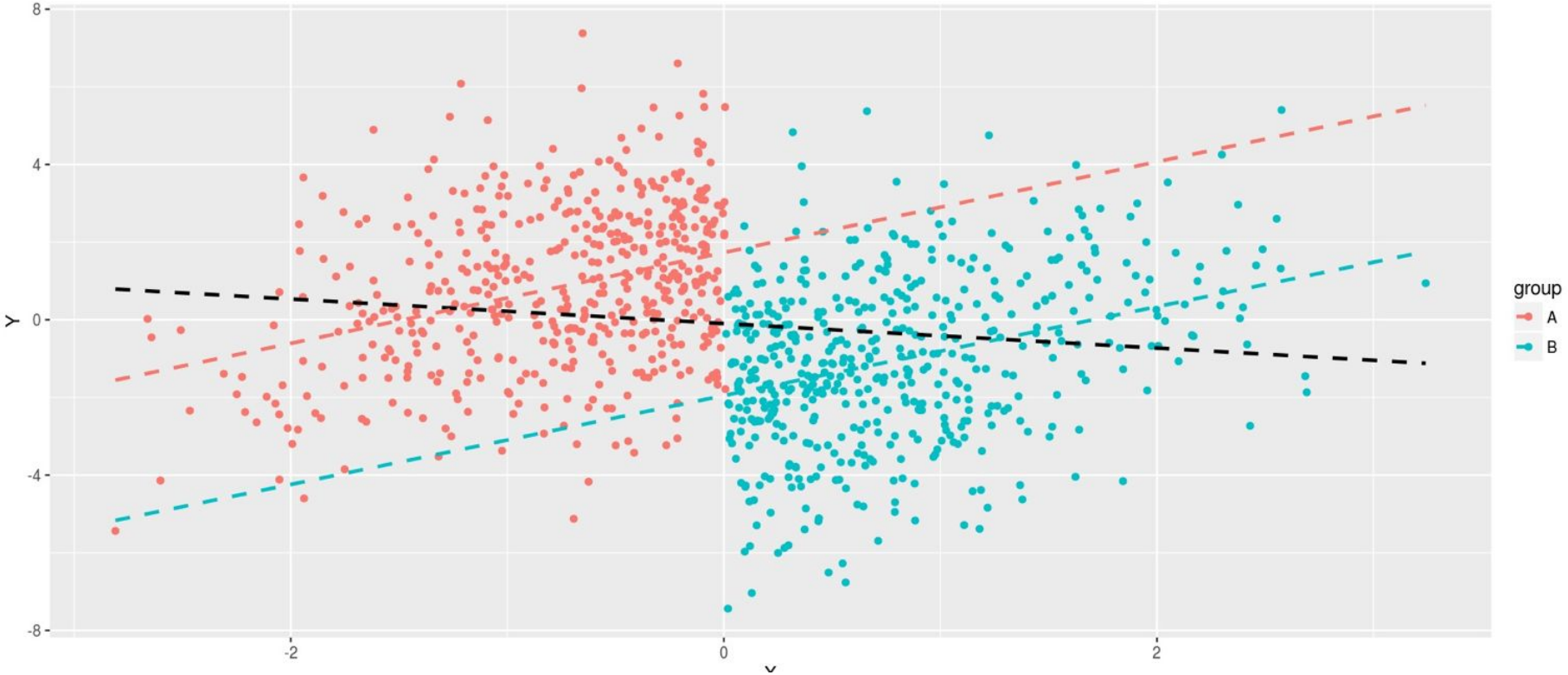
r = 0.702033103659 ; p = 0.000559678419262

Simpson’s Paradox



Source: Emil O. W. Kirkegaard

Simpson's Paradox



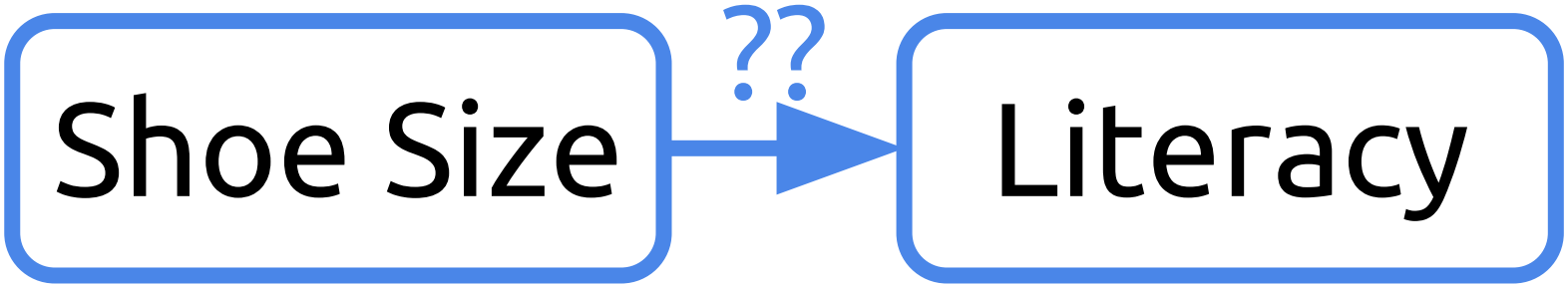
Source: Emil O. W. Kirkegaard

Subsampling and testing stability of your model's metrics help to determine how much of an effect outliers have on your results

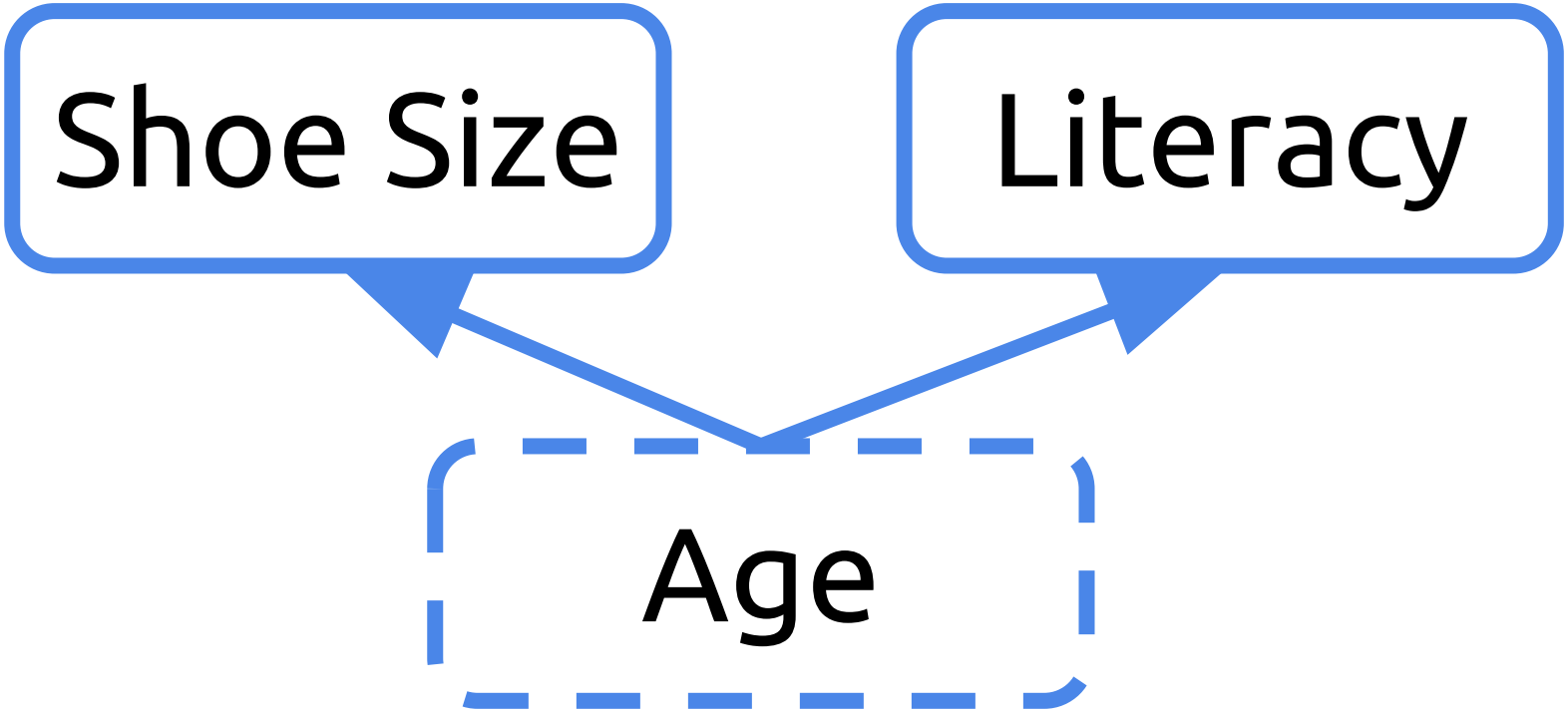
Confounding

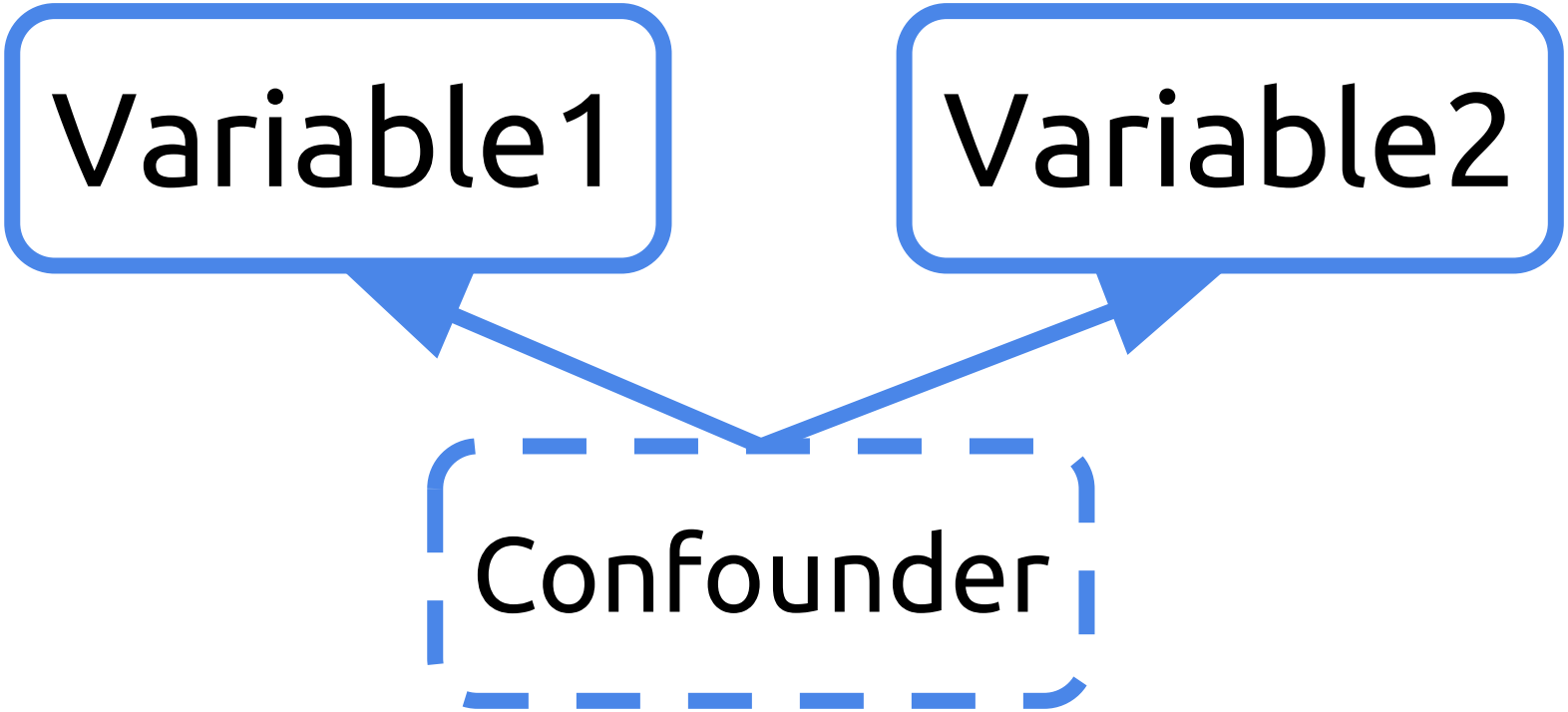
mixing up one thing with something else
so that the individual elements become difficult to distinguish

Small shoes
Not literate
Big shoes
Literate


Shoe Size
??
Literacy

Small shoes
Not literate
Young
Big shoes
Literate
Middle aged


Shoe Size
Literacy
Age

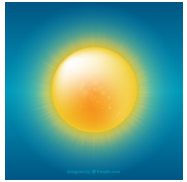

Variable1
Variable2
Confounder

popsicles → crime rate

popsicles → crime rate

temperature → popsicles

temperature → crime rate



You can plan ahead to avoid confounding and/or include confounders in your models to account for their role on the outcome variable

Ignoring confounders will lead you to draw incorrect conclusions from your analyses